



DEGREE PROJECT IN MATHEMATICS,
SECOND CYCLE, 30 CREDITS
STOCKHOLM, SWEDEN 2016

Evaluation of Missile Guidance and Autopilot through a 6 DOF Simulation Mode

ULF SEFASTSSON

Evaluation of Missile Guidance and Autopilot through a 6 DOF Simulation Mode

U L F S E F A S T S S O N

Master's Thesis in Systems Engineering (30 ECTS credits)
Degree Programme in Aerospace Engineering (120 credits)
Royal Institute of Technology year 2016
Supervisors at Swedish Defence University: Peter Bull
Supervisor at KTH: Per Engvist
Examiner: Per Engvist

TRITA-MAT-E 2016:38
ISRN-KTH/MAT/E--16/38--SE

Royal Institute of Technology
SCI School of Engineering Sciences

KTH SCI
SE-100 44 Stockholm, Sweden

URL: www.kth.se/sci

Abstract

Missile guidance and autopilot have been active fields of research since the second world war. There are lots of literature on the subjects, but the bulk of which are confined to overly simplified models, and therefore the publications of the methods applied to more realistic models are scarce.

In this report a nonlinear 6 DOF simulation model of a tail-controlled air-to-air missile is considered. Through several assumptions and simplifications a linearized approximation of the plant is obtained, which then is used in the implementation of 5 guidance laws and 2 autopilots. The guidance laws are all based on a linearized collision geometry, and the autopilots are based on model predictive control (MPC). Both autopilots use linear quadratic MPC (LQMPC), and one is more robust to modelling errors than the conventional LQMPC. The guidance laws and autopilots are then evaluated with respect to performance in terms of miss distance in 4 interception scenarios with a moving target.

The results show that the in this model the autopilots perform equally well, and that the guidance laws with more information about the target generally exhibit smaller miss distances, but at the cost of a considerably larger flight time for some scenarios.

The conclusions are that the simplifying assumptions in the modelling are legitimate and that the challenges of missile control probably does not lie in the guidance or autopilot, but rather in the target tracking. Therefore it is suggested that future work include measurement noise and process disturbances in the model.

Keywords: Missile Guidance, Proportional Navigation, Optimal Guidance, Differential Game Guidance, Model Predictive Control.

Utvärdering av missilstyrlagar och -automat med en 6 DOF simuleringsmodell

Sammanfattning

Det har forskats kring styrlagarna och styrautomaterna för robotar sedan andra världskrigets. Det finns mycket litteratur på områdena, men merparten av de publicerade resultaten behandlar enbart grovt förenklade modeller, och därför är tillgången på publikationer där metoderna applicerats i en mer realistisk modell begränsat.

I denna rapport behandlas en olinjär simuleringsmodell av en jaktrobot som styrs med stjärtfenor och har sex frihetsgrader. Genom en rad antaganden och förenklingar erhålls en linjäriserad modell av missilen, vilket sedan används för implementering av fem styrlagar och två styrautomater. Styrlagarna är alla baserade på en linjäriserad kollisionsgeometri och styrautomaterna är baserade på modellprediktiv styrning (MPC). Båda styrautomaterna använder linjärvadratisk MPC, där den ena påstås vara mer robust gentemot modellfel. Styrlagarna och -automaterna utvärderas ur ett prestandaperspektiv med fokus på bomavstånd i fyra realistiska genschjutningsscenarier med ett rörligt mål.

Resultaten visar att båda styrautomaterna presterar lika bra, och att de styrlagar med mer information om målets position/hastighet/acceleration generellt presterar bättre, men att de för vissa skjutfall får en väsentligt längre flygtid.

Slutsatserna är att förenkningarna och antagandena i linjäriseringen är välgrundade, och att utmaningarna i missilstyrning inte ligger i utformning av styrlag/-automat, utan förmodligen i målsökningen. Därför föreslås det slutligen att framtida arbete bl. a. inkluderar mätbrus och störningar.

Nyckelord: Missilstyrning, syftbäringsstyrning, optimal missilstyrning, spelteoretisk missilstyrning, modellprediktiv reglering.

Acknowledgements

I would first and foremost like to thank my supervisor Peter Bull and Björn Persson at the Swedish Defence University for their continuous support and constructive feedback in my work on this thesis.

Secondly, I want to thank my brother, Per Sefastsson, for his patience discussing the problem with me in times of adversity.

Lastly, I want to thank everyone at the military-technology division at the Swedish Defence University for their warm welcoming and integrating me in the group.

Stockholm, June 2016

Ulf Sefastsson

Contents

Abstract	ii
Sammanfattning	iii
Acknowledgements	iv
Nomenclature	1
1 Introduction	5
1.1 Background	5
1.2 Problem Formulation	5
1.3 Limitations	5
1.4 Previous Work	6
2 Missile and Target Models	7
2.1 Coordinate systems	7
2.2 Missile Model	7
2.3 Missile Dynamics	9
2.4 Linearized Model	12
3 Guidance	19
3.1 Classical Guidance Laws	19
3.2 Modern Guidance Laws	20
4 Autopilot and Model Predictive Control	29
4.1 Linear Quadratic MPC	30
4.2 Robust MPC	34
5 Simulations	37
5.1 Simulation Scenarios	37
6 Results	39
6.1 Autopilot	39
6.2 Guidance Laws	40

7	Discussion, Conclusions and Future Work	47
7.1	Discussion	47
7.2	Conclusions	47
7.3	Future Work	48
	Bibliography	49
	Appendices	51
A	Plots of Coefficients	53

Nomenclature

Unless otherwise stated the following nomenclature is used

α	Angle of attack
β	Sideslip angle
δ	rudder deflection angle
λ	Line-of-sight angle
$\bar{\mathbf{X}}$	State vector of linear state space model
ω	Rotational velocity
$\mathbf{A}$	A-matrix of linear state space model
$\mathbf{B}$	B-matrix of linear state space model
$\mathbf{C}$	C-matrix of linear state space model
$\mathbf{D}$	D-matrix of linear state space model
$\mathbf{f}^B$	Resultant of aerodynamic forces and thrust
$\mathbf{g}^N$	Gravitational acceleration expressed in N -frame
$\mathbf{I}$	Inertia tensor
$\mathbf{J}$	Cost function
$\mathbf{m}^B$	Moments acting on the missile expressed in the B -frame
$\mathbf{q}_B^N$	Rotational quaternion from B -frame to N -frame
$\mathbf{q}_N^B$	Rotational quaternion from N -frame to B -frame
$\mathbf{v}$	State-vector
ω_n	Natural frequency of rudder dynamics
$\otimes$	Quaternion multiplication

ϕ_{HE}	Heading error
$\times$	Vector product
$ \square $	Euclidian norm of the vector replaced by the box
ζ	Relative damping of rudder dynamics
a_M	Actual missile acceleration
a_M^c	Commanded missile acceleration
a_T	Actual target acceleration
C_B^N	Transformation matrix from B -frame to N -frame
C_N^B	Transformation matrix from N -frame to B -frame
CA	Axial force coefficient
Cl	Roll moment coefficient
Cm	Pitch moment coefficient
CN	Normal force coefficient
Cn	Yaw moment coefficient
CY	Side force coefficient
Ma	Mach number
N'	Effective navigation ratio or navigation constant
p	Roll velocity (angular velocity around x -axis)
q	Pitch velocity (angular velocity around y -axis)
q_{dyn}	Dynamic pressure
r	Yaw velocity (angular velocity around z -axis)
R_{TM}	Line-of-sight range
s	Frequency variable
s_{ref}	Reference area
T	Time constant
t	Time
t_f	Time until end of flight

T_s	Sampling time
t_{go}	Time to go
V_C	Closing velocity
V_M	Missile velocity
V_T	Target velocity
x	Position in x -direction
y	Position in y -direction
z	Position in z -direction
ZEM	Zero effort miss

Chapter 1

Introduction

1.1 Background

Simulations are widely used in the field of military technology as field experiments are costly, if not impossible as the number of parameters often are too large to control. Simulations also play a key role in the teaching of concepts in military technology.

As part of a greater project the Swedish Defense University (SEDU) have commissioned this master thesis project to design a control system for, and simulate an air-to-air missile that can be implemented in the FlamesTM framework.

There exists a lot of literature covering the guidance laws used in this report, but most of the published results is limited to overly simplified models. Therefore it is of interest to study how these results compare to a more realistic scenario.

1.2 Problem Formulation

Given a mathematical model describing the dynamics of an air-to-air missile, the guidance law and autopilot resulting in the missile intercepting a moving target are to be designed and evaluated with respect to robustness and performance. The performance measure used in this report is the miss distance. A total of 5 guidance laws and 2 autopilots are to be implemented.

The missile has 6 degrees of freedom (DOF) and is cruciform (i.e. symmetric around two axes) with 4 control surfaces located at the tail of its body.

1.3 Limitations

The limitations of the simulations are that no measurement noise or process disturbances are included in the model.

1.4 Previous Work

There are a number of previous theses on the subject of missile control. The most significant ones are [1] and [2], which both use model predictive control in combination with a few fundamental guidance laws, and [3] which implements $\mathcal{H}_\infty$ -control for the autopilot design.

Common to these theses is that the laws of motion including all forces and moments are known explicitly as smooth functions, which is not the case in this thesis.

Chapter 2

Missile and Target Models

In this chapter the mathematical models of the missile used in the simulation and control synthesis are introduced and derived.

2.1 Coordinate systems

Two cartesian coordinate systems are used:

North-East-Down has its origin in an arbitrary point on the surface of the earth with the x-axis pointing to the north, the y-axis pointing to the east and with the z-axis defined as positive pointing towards the center of the earth. Thus, the xy-plane is a tangential plane to the earth.

This coordinate system is denoted N .

Body is fixed with respect to the missile with the x-axis pointing in the forward direction of the missile (which need not coincide with the velocity vector), the y-axis pointing to the right and with the z-axis pointing downwards. The origin is located in the missiles center of mass. Note that the directions "downwards" and "right" do not have any physical meaning as the missile is cruciform and does not have a preferred orientation with respect to the N -system.

This coordinate system is denoted B , or body-fixed and is seen in Figure 2.1.

2.2 Missile Model

A schematic representation of the missile and the body fixed coordinate system can be seen in Figure 2.1. The missile is cruciform with four fins located at the tail. The missile has 6 degrees of freedom, which implies that,

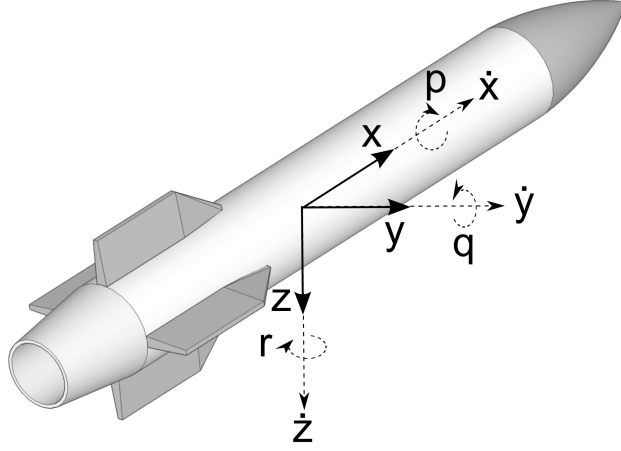


Figure 2.1: The missile and the body-fixed coordinate-system illustrated.

if the missiles position/orientation is not of interest, 6 states are required to capture the dynamics as shown in Table 2.1.

Table 2.1: States used in the missile model.	
Notation	Description
$\dot{x}$	Velocity in forward direction
$\dot{y}$	Velocity in sideways direction
$\dot{z}$	Velocity in downward direction
p	Roll angular velocity, rotation around x -axis
q	Pitch angular velocity, rotation around y -axis
r	Side slip angular velocity, rotation around z -axis

2.2.1 Model Assumptions

The assumptions for the missile simulations are

Assumption 1 - *The coordinates of the center of mass with respect to the B -frame are constant.*

Assumption 2 - *The moments of inertia with respect to the center of mass are constant.*

Assumption 3 - *The magnitude of the angle of attack and the sideslip angle are $\leq 20^\circ$.*

Assumption 3 is a hard constraint, rather than a simplification. The model used for simulations is not reliable whenever $|\alpha|, |\beta| > 20^\circ$.

2.2.2 Phases of Flight

The missile flight is characterized by three phases

Boost phase is characterized by its rapid acceleration to terminal speed.

Cruise phase is characterized by low to medium control action. The missile approaches the target, but is restrictive with large lateral accelerations to minimize induced drag and thus minimize the flight time.

Terminal phase is generally the last second(s) of flight. During this phase the permissible control gain is often chosen to be large in order to minimize miss distance.

In this report no division between the phases of flight is made, but the same control law (guidance and autopilot) is used throughout the missile flight. The flight time of the missile is 8 seconds where the boost phase is 4 seconds and accelerates the missile from Ma 0.8 to approximately Ma 3. The initial speed of Ma 0.8 corresponds to the missile being fired from an aircraft.

2.3 Missile Dynamics

In this section the equations describing the motion of the missile are presented. These are the ones used when simulating the missile. The dynamics used for control synthesis are developed in section 2.4.

2.3.1 Laws of Motion

The dynamics of the missile are governed by Newton-Eulers equations

$$\dot{\mathbf{x}}^N = C_B^N \mathbf{v}^B, \quad (2.1)$$

$$\dot{\mathbf{q}}_N^B = \frac{1}{2} \mathbf{q}_B^N \otimes \omega^B, \quad (2.2)$$

$$\dot{\mathbf{v}}^B = C_N^B \mathbf{g}^N + \frac{1}{m} \mathbf{f}^B - \omega^B \times \mathbf{v}^B, \quad (2.3)$$

$$\dot{\omega}^B = \mathbf{I}^{-1} (\mathbf{m}^B - \omega^B \times \mathbf{I} \omega^B). \quad (2.4)$$

(2.1) is saying that the the rate of change in position of the missile in the N -frame equals the transformation matrix C_B^N (transforming coordinates from B to N) multiplied by the missiles velocity expressed in B .

(2.2) is describing the rotational quaternion $\mathbf{q}_B^N$ rate of change. The quaternion is then related back to the transformation matrix C_B^N . The $\otimes$ is quaternion multiplication. The raison d'être of the quaternion is that it eliminates the singularities for large attitude angles emerging when differentiating Euler angles. In chapter 10 of [4] the explanation of the method of converting between transformation matrices and quaternions can be found.

(2.3) says that the linear acceleration of the missile in the Body-fixed frame B equals the gravity (which is positive in z-direction in coordinate system N) transformed to B through C_N^B plus the aerodynamic forces and thrust $\mathbf{f}^B$ divided by the mass minus the cross-product of the angular velocity and the velocity vectors.

(2.4) says that the angular acceleration of the missile in the Body-fixed frame B equals the inverse of the inertia tensor $\mathbf{I}$ times the moments from aerodynamics and thrust $\mathbf{m}^B$ minus the cross product of the angular velocity and the inertia tensor times the angular velocity. The inertia tensor in this model is a diagonal matrix

$$\mathbf{I} = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}, \quad (2.5)$$

and as the missile is cruciform it follows that $I_{yy} = I_{zz}$.

The resulting force $\mathbf{f}^B$ and moment $\mathbf{m}^B$ can be decomposed into one component due to aerodynamics and one due to thrust. The components from aerodynamics become

$$\mathbf{f}_A^B = s_{ref} q_{dyn} \begin{bmatrix} -CA \\ CY \\ -CN \end{bmatrix}, \quad (2.6)$$

$$\mathbf{m}_A^B = s_{ref} q_{dyn} \begin{bmatrix} Cl \\ Cm \\ Cn \end{bmatrix}. \quad (2.7)$$

CA , CY , CN , Cl , Cm and Cn are all coefficients dependent on α , β , Ma , δ_a , δ_e and δ_r . S_{ref} is the reference area, which is a missile specific parameter, and the dynamic pressure q_{dyn} is

$$q_{dyn} = \frac{\rho ||\mathbf{v}^B||^2}{2}. \quad (2.8)$$

The second component of $\mathbf{f}^B$ and $\mathbf{m}^B$ are denoted $\mathbf{f}_T^B$ and $\mathbf{m}_T^B$. They are piecewise constant and zero, respectively. This is because the thrust is acting in the forward direction of the missile through the center of gravity, thus causing no moments on the missile.

2.3.2 Coefficients

In Table 2.2 the coefficients introduced in (2.6) and (2.7) are described. In the simulation of the model the coefficients are interpolated from tabulated values for the current flight regime.

Table 2.2: Aerodynamic coefficients used in the missile model.

Notation	Description
$CA(\alpha, \beta, Ma, \delta_a, \delta_e, \delta_r)$	Axial force coefficient
$CN(\alpha, \beta, Ma, \delta_a, \delta_e, \delta_r)$	Normal force coefficient
$CY(\alpha, \beta, Ma, \delta_a, \delta_e, \delta_r)$	Side force coefficient
$Cl(\alpha, \beta, Ma, \delta_a, \delta_e, \delta_r)$	Roll moment coefficient
$Cm(\alpha, \beta, Ma, \delta_a, \delta_e, \delta_r)$	Pitch moment coefficient
$Cn(\alpha, \beta, Ma, \delta_a, \delta_e, \delta_r)$	Yaw moment coefficient

2.3.3 Actuators

The actuators of the missile are constituted of 4 fins located at the tail. The configuration with respect to the Body-fixed frame can be seen in Figure 2.2.

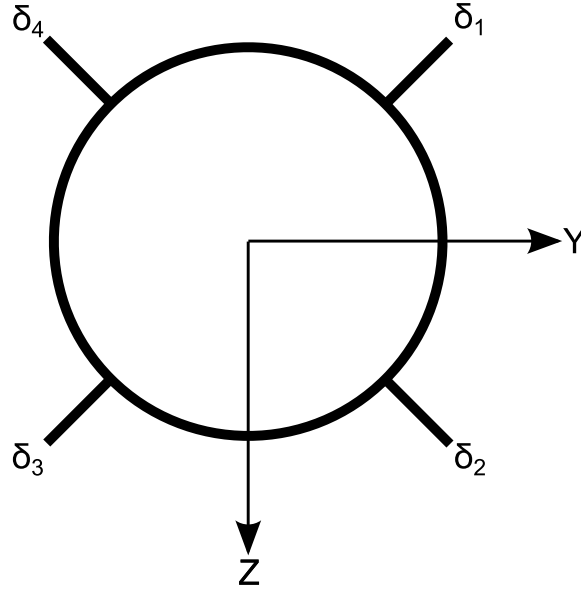


Figure 2.2: The fin configuration seen from the rear.

As the coefficients in (2.6) and (2.7) are functions of δ_a , δ_e and δ_r , the 4 input signals need to be transformed into 3 equivalent fin angles (δ_a = aileron, δ_e = elevator, δ_r = rudder). The relationship is given by

$$\begin{bmatrix} \delta_a \\ \delta_e \\ \delta_r \end{bmatrix} = \frac{1}{4} \begin{bmatrix} -1 & -1 & -1 & -1 \\ -1 & -1 & 1 & 1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \\ \delta_4 \end{bmatrix}. \quad (2.9)$$

The matrix in (2.9) can be pseudo-inverted, yielding

$$\begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \\ \delta_4 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 1 \\ -1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \delta_a \\ \delta_e \\ \delta_r \end{bmatrix}. \quad (2.10)$$

This implies that the control commands equally well can be expressed either in terms of the real or the equivalent fin angles, as the relationship between the two is a static one.

Further, the dynamics for each rudder is modeled as a second order system. The transfer function relating the commanded deflection δ^c to the actual deflection δ is

$$\delta = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \delta^c, \quad (2.11)$$

where ω_n is the eigenfrequency and ζ is the relative damping of the system. On state-space form (2.11) can be written

$$\begin{bmatrix} \dot{\delta} \\ \ddot{\delta} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix} \begin{bmatrix} \delta \\ \dot{\delta} \end{bmatrix} + \begin{bmatrix} 0 \\ \omega_n^2 \end{bmatrix} \delta^c. \quad (2.12)$$

2.4 Linearized Model

In this section the method used in the linearizing approximation of the missile model is first described. Then the nonlinearities of the missile model are explained, along with the assumptions allowing for a linearized model. Lastly, the resulting linearized state space model is presented.

2.4.1 Gain Scheduling

For a nonlinear plant designed to work in the vicinity of one operating point, a single linearization is sufficient. In applications where the operating point and/or the system dynamics vary greatly with time (though not explicitly as functions of time), several linearizations can be derived and then any working point between the points for which an explicit linearization has been calculated is interpolated. This is called *gain scheduling*, and the key assumption of the concept is that the parameters used for scheduling are varying much slower than the states and/or the inputs of the system.

This approach to the control problem is widely used in aerospace applications. For example, autopilots are often constituted of linear combinations of different controllers depending on Mach-number and altitude. An illustration of this concept is shown in Figure 2.3.

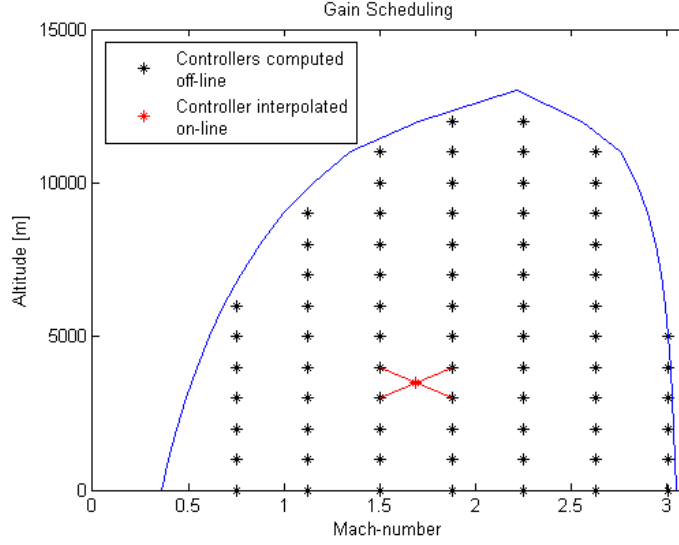


Figure 2.3: A schematic illustration of gain scheduling. The blue lines confining the black asterisks represents the flight envelope. The black asterisks represent controllers computed off-line. In this example the altitude is 3500 m and $Ma \approx 1.7$, which yields the controller resulting from interpolation of the four nearest controllers computed off-line.

2.4.2 Aerodynamic Coefficients

In Appendix A 9 plots of the coefficients are shown, and it can be seen that they are nonlinear over the flight envelope. However, if the Mach number is constant, the surfaces found in Appendix A are reduced to 2-D plots. This leads to the first assumption

Assumption 4 - *The Mach number is constant.*

Assumption 4 is valid for the cruise and terminal phase of the missile flight. From Assumption 4 it also follows that the forward velocity (in the B -frame) can be eliminated from the state vector, as it is deterministically given and not affected by inputs or any other state. Hence, CA does not affect the linearized missile model.

In Figure 2.4 the coefficient CY is plotted as a function of δ_r for different δ_a and δ_e (and with $Ma = 2.6$). CN , Cl , Cm and Cn exhibit the same behavior both with respect to fin angles and the angle of attack/sideslip angle, and are found in Appendix A. Therefore, the coefficients CY , CN , Cl , Cm and Cn are decomposed into two parts, one due to δ_e/δ_r and one due to α/β . For CY this can be written

$$CY = CY_{\delta_r} + CY_{\beta}, \quad (2.13)$$

and the same goes for CN , Cl , Cm and Cn .

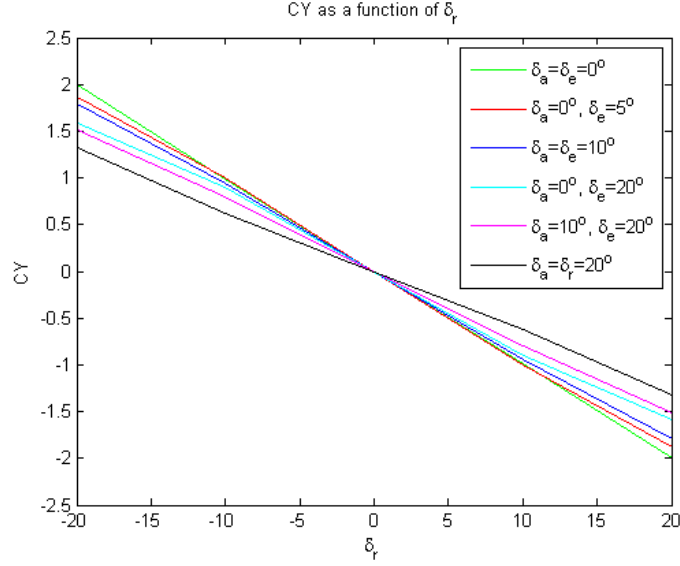


Figure 2.4: CY as a function of rudder deflection angle δ_r ($\alpha = \beta = 0$ and various combinations of δ_a and δ_e).

2.4.3 The Linear Acceleration

(2.3) is nonlinear in $\dot{x}$, $\dot{y}$, $\dot{z}$, p , q and r according to

$$\omega^B \times \mathbf{v}^B = \begin{bmatrix} \dot{z}q - \dot{y}r \\ \dot{x}r - \dot{z}p \\ \dot{y}p - \dot{x}r \end{bmatrix}. \quad (2.14)$$

As the velocity in the x -direction is assumed constant ($\dot{x} = V$), and if $\dot{y}$ and $\dot{z}$ are small, (2.14) becomes

$$\omega^B \times \mathbf{v}^B \approx \begin{bmatrix} 0 \\ Vr \\ -Vq \end{bmatrix}. \quad (2.15)$$

Further, the angle of attack and sideslip angle can then with the same reasoning be expressed as

$$\alpha \approx \frac{\dot{z}}{V}, \quad (2.16)$$

$$\beta \approx \frac{\dot{y}}{V}. \quad (2.17)$$

This is an approximation made which yields Assumption 5.

Assumption 5 - The angle of attack α and sideslip angle β are small.

It must anyways hold that $|\alpha|, |\beta| \leq 20^\circ$ from assumption 3.

Also, the first term of (2.3) needs to be addressed. This is the component due to gravity, which has constant magnitude but a direction depending on the missiles orientation. The component due to gravity is approximately 13 % of the total force acting on the missile during the cruise- and terminal phase depending on the simulation scenario.

Assumption 6 - *The gravitational force acting on the missile is neglected.*

Assumption 6 is not justified as 13 % is not negligible, but it is an assumption necessary for the control synthesis. The acceleration due to gravity is assumed to be measurable, and is compensated for by adding a term to the reference signal (see Chapter 3).

2.4.4 The Angular Acceleration

The second component of (2.3) is nonlinear in p , q and r according to

$$\mathbf{I}^{-1}\omega^B \times \mathbf{I}\omega^B = \begin{bmatrix} \frac{I_{zz}-I_{yy}}{I_{xx}}qr \\ \frac{I_{xx}-I_{zz}}{I_{yy}}pr \\ \frac{I_{yy}-I_{xx}}{I_{zz}}pq \end{bmatrix}. \quad (2.18)$$

As the missile is not supposed to spin, but keep a rather steady roll angle, the roll angle and its derivative are eliminated from the state vector. Instead they are used as scheduling parameters.

Assumption 7 - *The roll angle velocity is small and varying slowly.*

As δ_a only affects p , the roll angle velocity can be driven to zero by appropriate choice of δ_a , and so assumption 7 is valid.

2.4.5 Linearized State Space Model

The output of the linearized model is accelerations in y and z directions.

With (2.1)-(2.4), and Assumptions 4-7 the linearized state space model becomes

$$\begin{aligned} \dot{\mathbf{X}} &= \mathbf{A}\mathbf{X} + \mathbf{B}\mathbf{u}, \\ \mathbf{Y} &= \mathbf{C}\mathbf{X}. \end{aligned}$$

The state vector is

$$\mathbf{X} = [\dot{y} \quad \dot{z} \quad q \quad r \quad \delta_e \quad \delta_r \quad \dot{\delta}_e \quad \dot{\delta}_r]^T, \quad (2.19)$$

the input is

$$\mathbf{u} = \begin{bmatrix} \delta_e^c \\ \delta_r^c \end{bmatrix}, \quad (2.20)$$

and the matrices **A**, **B** and **C** are

$$\mathbf{A} = \begin{bmatrix} q_{dyn}s_{ref}\frac{C_{Y,a}}{mV} & p & 0 & -V & 0 & q_{dyn}s_{ref}\frac{C_{Y,\delta}}{m} & 0 & 0 \\ p & -q_{dyn}s_{ref}\frac{C_{N,a}}{mV} & V & 0 & q_{dyn}s_{ref}\frac{C_{N,\delta}}{m} & 0 & 0 & 0 \\ 0 & q_{dyn}s_{ref}d_{ref}\frac{C_{m,a}}{I_{yy}V} & 0 & \frac{I_{zz}-I_{xx}}{I_{yy}}p & q_{dyn}s_{ref}d_{ref}\frac{C_{m,\delta}}{I_{yy}} & 0 & 0 & 0 \\ q_{dyn}s_{ref}d_{ref}\frac{C_{n,a}}{I_{zz}V} & 0 & \frac{I_{yy}-I_{xx}}{I_{zz}}p & 0 & 0 & q_{dyn}s_{ref}d_{ref}\frac{C_{n,\delta}}{I_{zz}} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -\omega_n^2 & 0 & -2\zeta\omega_n & 0 \\ 0 & 0 & 0 & 0 & 0 & -\omega_n^2 & 0 & -2\zeta\omega_n \end{bmatrix},$$

$$\mathbf{B} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \omega_n^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \omega_n^2 \end{bmatrix}^T,$$

$$\mathbf{C} = \begin{bmatrix} q_{dyn}s_{ref}\frac{C_{Y,a}}{mV} & 0 & 0 & 0 & 0 & q_{dyn}s_{ref}\frac{C_{Y,\delta}}{m} & 0 & 0 \\ 0 & -q_{dyn}s_{ref}\frac{C_{N,a}}{mV} & 0 & 0 & q_{dyn}s_{ref}\frac{C_{N,\delta}}{m} & 0 & 0 & 0 \end{bmatrix}.$$

At every sampling instance the **A**, **B** and **C** matrices are updated based on information about ρ and Ma . The rudder angles are not used for gain scheduling as they vary almost as fast as the control signals.

Scheduling Parameters

To obtain a linear system it is necessary to schedule on ρ (which is depending on altitude) and Ma . Then the aerodynamic forces and moments in (2.6) and (2.7) can be locally approximated as linear functions of α , β and the inputs δ_a , δ_e and δ_r . The various coefficients dependency on Ma and fin deflection angle are illustrated in Figure A.2, A.3, A.4 and A.5.

Since the missile is designed to reach its terminal velocity during the first four seconds, $Ma \approx 2.8$ during the rest of the flight time. This is a convenient simplification since it can be seen in Figure A.6 that approximating CA by a linear function separable in δ_a , δ_e , δ_r is not viable.

2.4.6 Pairing Input and Output Signals

In the **A**-matrix it can be seen that if $p = 0$ the two inputs and outputs are perfectly decoupled, and the greater p the greater the cross coupling becomes. Therefore the pairing of inputs and outputs is obvious for small p .

In Figure 2.5 the step response of the transfer function from u_1 to y_2 is shown. The same behaviour is obtained when plotting the step response from u_2 to y_1 . The fact that the output initially becomes negative before it reverses and becomes positive implies that the system is non-minimum phase, i.e. the system has right half plane zeros. In Chapter 4 the control method handling the non-minimum phase behaviour and signal pairing is explained.

2.4.7 ZOH-sampling of the Continuous Time Model

The autopilot operates in discrete time, and therefore the system used for designing and implementing the control law must be expressed in discrete

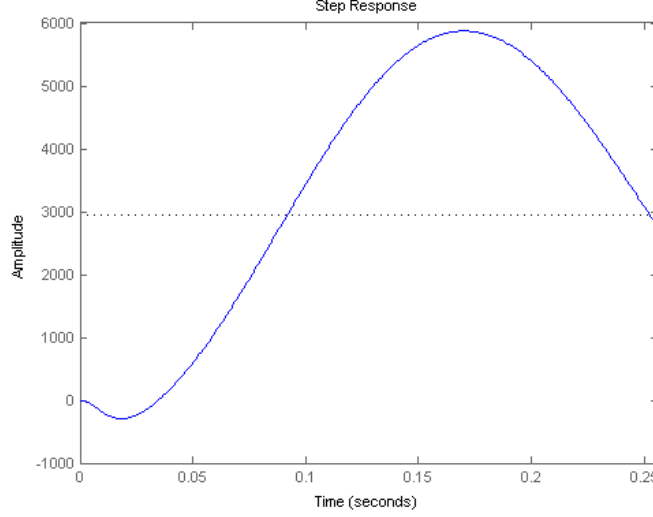


Figure 2.5: Example of a step response from u_1 to y_2 for a typical flight regime. The greater p is the greater the cross coupling between $u_1 \rightarrow y_1$ or $u_2 \rightarrow y_2$. For $p = 0$ there is no cross coupling.

time. In [5] it can be found that the difference equation

$$\mathbf{X}(kT_s + T_s) = e^{AT_s} \mathbf{X}(kT_s) + \int_0^{T_s} e^{As} ds \mathbf{u}(kT_s), \quad (2.21)$$

corresponds to the zero-order hold (ZOH) sampling of the continuous time system

$$\dot{\mathbf{X}} = \mathbf{A}\mathbf{X} + \mathbf{B}\mathbf{u}, \quad (2.22)$$

with sampling time T_s . ZOH means that the control signal $\mathbf{u}$ is held constant between sampling instances. In Matlab this is achieved using the built in `c2d`-function.

2.4.8 Sampling Time

When discretizing a continuous linear system a rule-of-thumb is to choose a sampling rate such that 6-10 samples are obtained per rise time. Plugging in $Ma = 2.6$, altitude 1000 m and $\beta = \alpha = 0$ the step response for the linearized system is evaluated, and can be seen in Figure 2.5. Graphically it is seen that the rise time is ≈ 0.04 seconds, which by the rule-of-thumb gives a maximum sampling time of $T_s = 0.0067$ seconds.

It is desired to keep the sampling time as large as possible, to minimize computational effort. From [1] and [2] a sampling time for similar missile models of $T_s = 0.01$ s has been used with success, and hence the sampling

time used is $T_s = 0.01$ s. (For the simulation scenarios in Chapter 5 it is found that a sampling time of up to 0.05 s yields satisfactory results).

Chapter 3

Guidance

The missile control system is often decomposed into tracking, navigation, guidance and autopilot. The tracking block measures and estimates the position, velocity and acceleration of the target and the navigation block determines the position, velocity, acceleration and attitude of the missile. The guidance system receives information about the missile and target positions, velocities and accelerations, based on which it computes commanded accelerations in the missiles y and z-directions. The guidance system then forwards the commanded accelerations to the autopilot, which tries to actualize the desired accelerations. This is seen in Figure 3.1.

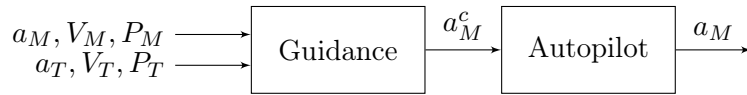


Figure 3.1: The input used in the Guidance-block varies depending on the guidance law used. P =position, V =velocity, a =acceleration. Subscript M for "missile" and T for "target". Superscript c for "commanded".

3.1 Classical Guidance Laws

3.1.1 Pure Pursuit Guidance

Pure pursuit is the oldest and most primitive of all recognized guidance schemes. Though it is questionable if it is ever used in any real applications, it is implemented in this report for completeness.

The principle is simple. The objective of the missile is to keep its velocity vector directed towards the target at all times. The commanded acceleration is thus proportional to the heading error, not taking into account that the target is moving. Mathematically

$$a_M^c = N' \phi_{HE}, \quad (3.1)$$

where N' is the navigation constant and ϕ_{HE} is the heading error. In Figure 3.2 the heading error is

$$\phi_{HE} = \gamma - \lambda. \quad (3.2)$$

3.2 Modern Guidance Laws

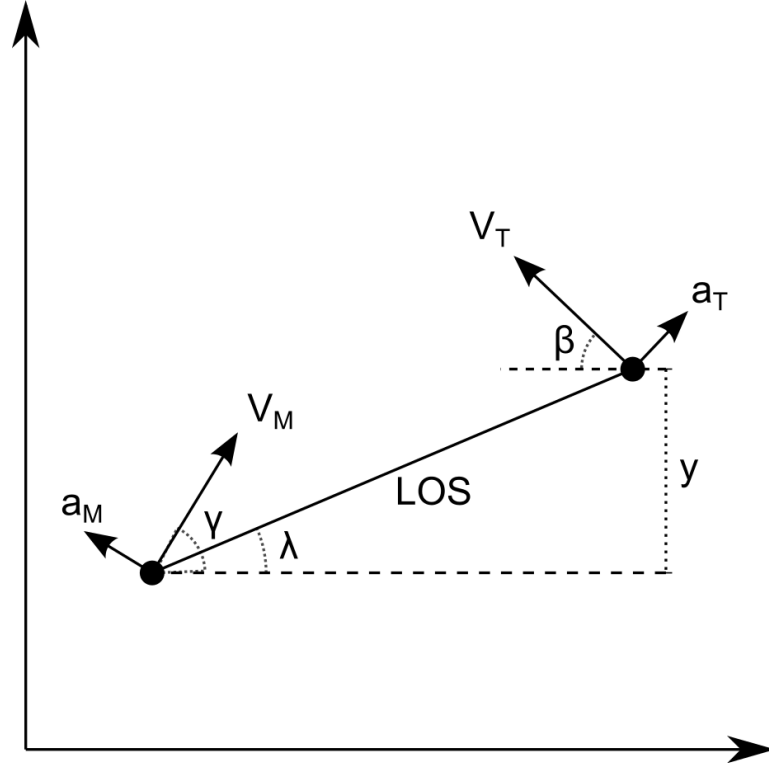


Figure 3.2: Geometry in 2-D for the interception. M denotes missile and T target. LOS stands for "line of sight", λ is then the LOS-angle. y is the relative separation. V and a are velocity and acceleration respectively.

In Figure 3.2 the collision plane with the missile and the target is shown. y is called the relative separation, and if $\gamma = \lambda$ (it is assumed that the two angles are at least approximately equal) the relative acceleration becomes

$$\ddot{y} = a_T \cos \beta - a_M \cos \lambda. \quad (3.3)$$

If the angles λ and β are small (3.3) can be approximated by

$$\ddot{y} = a_T - a_M, \quad (3.4)$$

and if λ is small and $\beta \approx \pi$

$$\ddot{y} = a_T + a_M. \quad (3.5)$$

The LOS-angle λ can under the same assumption be approximated by

$$\lambda = \frac{y}{R_{TM}}, \quad (3.6)$$

where R_{TM} is the LOS range.

The closing velocity, V_c , is introduced

$$V_c = V_M \cos \lambda + V_T \cos \beta. \quad (3.7)$$

For small angles λ and β the missile and target are almost facing each other head on, and the closing velocity is defined as

$$V_c = V_M + V_T. \quad (3.8)$$

If λ is small and $\beta \approx \pi$, the target and the missile are approximately in perfect pursuit and the closing velocity is defined as

$$V_c = V_M - V_T. \quad (3.9)$$

From the assumption two new concepts are introduced. The time-to-go, t_{go} , is the time until end of flight, where V_c becomes negative, and is defined by

$$t_{go} = t_f - t = \frac{R_{TM}}{V_c}, \quad (3.10)$$

where t_f is the final time.

The miss distance is then the relative separation at the final time t_f

$$\text{Miss} = y(t_f). \quad (3.11)$$

If the target continues along its path and the missiles acceleration is zero, the Miss is called the Zero Effort Miss (*ZEM*).

3.2.1 Proportional Navigation

The dividing line between classical and modern missile guidance is often said to be proportional navigation (PN). The advantage of PN-guidance is its simplicity. The missile only requires information about the closing velocity and the LOS-angle rate of change, which sets lower requirements on the target tracking than if target acceleration was to be used.

The principle of PN-guidance is simple. If the LOS-angle is constant and the closing velocity is positive collision will take place. a_T is assumed to be zero.

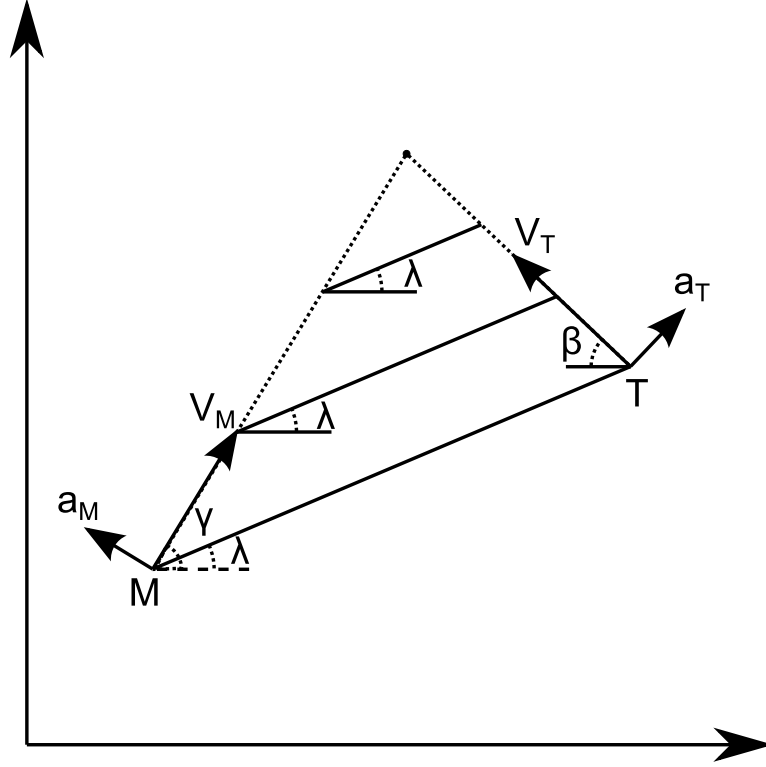


Figure 3.3: Geometry in 2-D for PN-guidance. The objective of the missile is to keep the LOS-rate equal to zero.

There are different versions of PN-guidance, and the one used in this report is called true proportional navigation. The commanded acceleration is a navigation constant times the product of the closing velocity and the LOS-angle rate of change. Thus,

$$a_M^c = N' V_c \dot{\lambda}. \quad (3.12)$$

The navigation constant N' is usually chosen to lie between 3 and 5. The principle is shown in Figure 3.3.

From Figure 3.3 it can be deduced that (3.12) alternatively can be expressed as

$$a_M^c = \frac{N'}{t_{go}^2} [y + \dot{y} t_{go}], \quad (3.13)$$

where the bracketed term in (3.13) is the ZEM .

3.2.2 Augmented Proportional Navigation

Augmented Proportional Navigation (APN) is PN assuming the target is performing a maneuver with constant lateral acceleration. Therefore an extra

term is augmented to the PN-law, compensating for the constant turn.

The guidance law is defined by

$$a_M^c = N' V_c \dot{\lambda} + \frac{N' a_T}{2}. \quad (3.14)$$

The advantage of this guidance law is that if the target acceleration is estimated correctly the commanded acceleration is greater earlier in the flight, and decreases as t_{go} decreases (for constant target maneuvers), compared to PN where the commanded acceleration increases monotonically as t_{go} decreases. Analogous to the case of PN-guidance, (3.14) can be expressed as

$$a_M^c = \frac{N'}{t_{go}^2} \left[y + \dot{y} t_{go} + \frac{1}{2} a_T t_{go}^2 \right], \quad (3.15)$$

where once again the bracketed term in (3.15) is the *ZEM*.

3.2.3 Optimal Guidance Law

In this section the optimal guidance law (OGL) is derived. The result is important as it will be seen that PN and APN merely are special cases of OGL [6].

Before one can embark such task the notion of optimality must be defined. The objective of any guidance law is to minimize the miss distance, but this is achieved by PN- and APN-guidance as well. A natural choice for the objective could then instead be to minimize the flight time or the control effort. In this guidance scheme the square of the control effort integrated over the flight time is minimized according to

$$\min \int_0^{t_f} (a_M^c(t))^2 dt, \quad (3.16)$$

with the boundary constraint

$$y(t_f) = 0. \quad (3.17)$$

The assumption of constant target lateral acceleration still applies, but OGL has the advantage over APN that it can take time constants of the missile dynamics into account. Consider the following transfer function from commanded acceleration a_M^c to actual acceleration a_M

$$a_M = \frac{1}{1 + sT_M} a_M^c, \quad (3.18)$$

where T_M is the missile time constant.

The dynamics of the linearized collision plane in Figure 3.2 becomes

$$\begin{bmatrix} \dot{y} \\ \ddot{y} \\ \dot{a}_T \\ \dot{a}_M \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{-1}{T_M} \end{bmatrix} \begin{bmatrix} y \\ \dot{y} \\ a_T \\ a_M \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{T_M} \end{bmatrix} a_M^c. \quad (3.19)$$

A fundamental matrix to (3.19) is

$$\Psi(t) = \begin{bmatrix} 1 & t & \frac{t^2}{2} & -tT_M + T_M^2(1 - e^{-t/T_M}) \\ 0 & 1 & t & -T_M(1 - e^{-t/T_M}) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{-t/T_M} \end{bmatrix}. \quad (3.20)$$

The optimal control problem defined by (3.16), (3.17) and (3.20) cannot be solved analytically through traditional optimal control techniques such as Pontryagin's Minimum Principle. The solution follows.

Consider the first row of the fundamental matrix in (3.20). It gives

$$y(t_f) = y(t) + \dot{y}(t)(t_f - t) + \frac{1}{2}a_T(t_f - t)^2 + a_M(T_M^2(1 - e^{-(t_f - t)/T_M} - (t_f - t)T_M) - \frac{1}{T_M} \int_t^{t_f} (t_f - \tau)a_M^c(\tau)d\tau). \quad (3.21)$$

If

$$f_1(t_f - t) = y(t) + \dot{y}(t)(t_f - t) + \frac{1}{2}a_T(t_f - t)^2 + a_M(T_M^2(1 - e^{-(t_f - t)/T_M} - (t_f - t)T_M), \quad (3.22)$$

and

$$h_1(t_f - t) = (t_f - t), \quad (3.23)$$

then (3.21) can be rewritten as

$$y(t_f) = f_1(t_f - t) - \int_t^{t_f} h_1(t_f - \tau)a_M^c(\tau)d\tau. \quad (3.24)$$

As $y(t_f) = 0$,

$$f_1(t_f - t) = \int_t^{t_f} h_1(t_f - \tau)a_M^c(\tau)d\tau. \quad (3.25)$$

Now, applying Schwartz inequality [7] to (3.25) yields

$$f_1^2(t_f - t) \leq \int_t^{t_f} h_1^2(t_f - \tau)d\tau \int_t^{t_f} (a_M^c(\tau))^2 d\tau, \quad (3.26)$$

or equivalently

$$\int_t^{t_f} (a_M^c(\tau))^2 d\tau \geq \frac{f_1^2(t_f - t)}{\int_t^{t_f} h_1^2(t_f - \tau)d\tau}. \quad (3.27)$$

Now the left hand side of (3.27) is the objective function, and for it to be minimized the equality must hold. According to Schwartz inequality that holds when

$$a_M^c(t) = \left[\frac{f_1(t_f - t)}{\int_t^{t_f} h_1^2(t_f - \tau)d\tau} \right] h_1(t_f - t). \quad (3.28)$$

After some algebra the optimal guidance law can finally becomes

$$\begin{aligned}
a_M^c &= \frac{N'}{t_{go}^2} \left[y + \dot{y}t_{go} + \frac{1}{2}a_T t_{go}^2 - a_M T_M^2 (e^{-x} + x - 1) \right], \\
N' &= \frac{6x^2 (e^{-x} - 1 + x)}{2x^3 + 3 + 6x - 6x^2 - 12xe^{-x} - 3e^{-2x}}, \\
x &= \frac{t_{go}}{T_M}.
\end{aligned} \tag{3.29}$$

The result in (3.29) is interesting when comparing to PN and APN. First, if there is no time constant for the missile dynamics, then $T_M \rightarrow 0$ and $N' \rightarrow 3$, $\forall t_{go} > 0$. Secondly, for $T_M = 0$ the commanded acceleration becomes

$$a_M^c = \frac{3}{t_{go}^2} \left[y + \dot{y}t_{go} + \frac{1}{2}a_T t_{go}^2 \right]. \tag{3.30}$$

(3.30) is equivalent to APN-guidance law in (3.15) with $N' = 3$, and if $a_T = 0$ it is equivalent to the PN-guidance law in (3.13).

3.2.4 Differential Game Guidance

First introduced in [8], the interception problem is formulated as a pursuit-evasion zero sum game. The advantage of this approach compared to OGL is that it does not require any information about the acceleration of the target, but only about its minimum/maximum admissible acceleration. Further, the differential game guidance (DG-guidance) allows the target to also be approximated by a first order system.

The pursuit-evasion game is according to [9] defined by:

1. An admissible game space
2. Differential equations governing the game dynamics
3. Initial conditions
4. Admissible control sets of the players
5. Conditions of game termination
6. A cost function J
7. The players' roles (minimizer vs. maximizer)
8. The information pattern of the game

The first 4 elements are trivially given as the game space is the collision plane and the dynamics are given together with initial conditions and bounds on

controls. The condition of termination is when the closing velocity switches sign. In this report the cost function used is

$$J(u, v) = \frac{1}{2}Z^2(t_f) + \frac{1}{2} \int_t^{t_f} \alpha u^2(\tau) - \beta v^2(\tau) d\tau, \quad (3.31)$$

where $u(t)$ is the purser (missile) control, $v(t)$ is the evader (target) control and $Z(t)$ is the ZEM , and so $Z(t_f)$ is the miss distance. Thus, the missile is the minimizer and the target is the maximizer. Both players are assumed to have perfect information about the game and all its elements.

The dynamics of the system take the following form

$$\dot{\mathbf{X}} = \mathbf{A}\mathbf{X} + \mathbf{B}u + \mathbf{C}v. \quad (3.32)$$

With the same state-vector $\mathbf{X}$ as in the case of OGL, (3.32) becomes

$$\begin{bmatrix} \dot{y} \\ \ddot{y} \\ \dot{a}_T \\ \dot{a}_M \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -\frac{1}{T_T} & 0 \\ 0 & 0 & 0 & -\frac{1}{T_M} \end{bmatrix} \begin{bmatrix} y \\ \dot{y} \\ a_T \\ a_M \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{T_M} \end{bmatrix} a_M^c + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{T_T} \\ 0 \end{bmatrix} a_T^c, \quad (3.33)$$

where T_M is the time constant of the missile as in OGL, and T_T is that of the target.

With $\Phi(t_f, t)$ as the transition matrix $Z(t)$ becomes

$$Z(t) = \mathbf{D}\Phi(t_f, t)\mathbf{X}(t). \quad (3.34)$$

Applying the controls $u(t)$ and $v(t)$ yields the miss distance at the final time

$$Z(t_f) = Z(t) + \int_t^{t_f} \tilde{B}(t_f, t)u(t) + \tilde{C}(t_f, t)v(t)dt, \quad (3.35)$$

where

$$\tilde{B}(t_f, t) = \mathbf{D}\Phi(t_f, t)\mathbf{B}, \quad (3.36)$$

$$\tilde{C}(t_f, t) = \mathbf{D}\Phi(t_f, t)\mathbf{C}, \quad (3.37)$$

and

$$\mathbf{D} = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}. \quad (3.38)$$

If the controls u and v are normalized, $\tilde{B}(t_f, t)$ and $\tilde{C}(t_f, t)$ become

$$\tilde{B}(t_f, t) = -a_{M,max}T_M (e^{-x} + x - 1), \quad (3.39)$$

$$\tilde{C}(t_f, t) = a_{T,max} (e^{-x/\epsilon} + x/\epsilon - 1), \quad (3.40)$$

where $x = t_{go}/T_M$ as in the case of OGL, and $\epsilon = T_T/T_M$.

Details on how to solve this type of problems can be found in [8], and therefore some of the steps in the derivation are omitted in this report. The Hamiltonian for the game with cost function (3.31) and $Z(t_f)$ defined by (3.35) is

$$H(Z, u, v, \lambda_Z) = \frac{1}{2}(\alpha u^2 - \beta v^2) + \lambda_Z \left(\tilde{B}(t_f, t)u + \tilde{C}(t_f, t)v \right). \quad (3.41)$$

λ_Z is the adjoint variable, which must obey

$$\dot{\lambda}_Z = -\frac{\partial H}{\partial Z} = 0, \quad \lambda_Z(t_f) = \frac{\partial \phi}{\partial Z} = Z(t_f), \quad (3.42)$$

$$\implies \lambda_Z(t) = Z(t_f). \quad (3.43)$$

Since the objective function is separable in u and v it makes no difference in which order the minimization/maximization occurs. Minimizing/maximizing pointwise yields

$$u^*(t) = \arg \min_u \{H\} = -\frac{\tilde{B}(t_f, t)}{\alpha} Z(t_f), \quad (3.44)$$

$$v^*(t) = \arg \max_v \{H\} = \frac{\tilde{C}(t_f, t)}{\beta} Z(t_f). \quad (3.45)$$

Note that though Pontryagin's principle only gives sufficient conditions for optimality it can easily be seen that $\frac{\partial^2 H}{\partial u^2} > 0$ and $\frac{\partial^2 H}{\partial v^2} < 0$, and so the candidate solution is optimal (saddle point). After some tedious steps that can be found in [9], and after letting α and β tend to zero (thus penalizing miss distance rather than control), one finally arrives at the optimal strategies

$$u^*(t) = \frac{-\tilde{B}(t_f, t)}{\int_t^{t_f} \tilde{B}^2(t_f, \tau) d\tau - \sigma \int_t^{t_f} \tilde{C}^2(t_f, \tau) d\tau}, \quad (3.46)$$

$$v^*(t) = \frac{\tilde{C}(t_f, t)}{\frac{1}{\sigma} \int_t^{t_f} \tilde{B}^2(t_f, \tau) d\tau - \int_t^{t_f} \tilde{C}^2(t_f, \tau) d\tau}, \quad (3.47)$$

where $\sigma = \frac{\alpha}{\beta}$.

Now, letting the target be ideal ($T_T = 0$) and after some algebra, the DG-guidance law becomes

$$a_M^c = u^*(t) = \frac{N}{t_{go}^2} [y + \dot{y}t_{go} - a_M T_M^2 (e^{-x} + x - 1)], \quad (3.48)$$

where the effective navigation ratio is

$$N' = \frac{6x^2 (e^{-x} + x - 1)}{2 \left(1 - \left(\frac{\sigma}{\mu} \right)^2 \right) x^3 + 3 - 6x^2 + 6x - 12xe^{-x} - 3e^{-2x}} \quad (3.49)$$

and

$$\mu = \frac{a_{M, max}}{a_{T, max}}. \quad (3.50)$$

Chapter 4

Autopilot and Model Predictive Control

The purpose of the autopilot is to act as a servo, driving the actual acceleration to the desired acceleration commanded by the guidance law as seen in Figure 3.1. As the control signals (fin deflections), output (load factor) and states (angle of attack/sideslip angle) are constrained, this must be taken into consideration when designing and implementing a control algorithm. Further, the non-minimum phase behaviour of the system along with the model-plant mismatch can be hard to handle using classic control methods such as PID-design. Model predictive control (MPC) is an optimization-based control method that takes the non-minimum phase behaviour into consideration and can also handle constraints on output, states and control signals.

MPC works in discrete time and computes the optimal control signal on-line every sampling instance. The control problem is formulated as an optimization problem on the form

$$\begin{aligned} \min_{\mathbf{U}} \quad & J(\mathbf{x}_t, \mathbf{U}) = \Psi(\mathbf{x}_{t+N}) + \sum_{i=t}^{t+N-1} \Phi(\mathbf{x}_i, \mathbf{u}_i) \\ \text{s.t.} \quad & \mathbf{x}_{i+1} = f(\mathbf{x}_i, \mathbf{u}_i), \\ & \mathbf{x}(t) = \mathbf{x}_t, \\ & \mathbf{U} = [\mathbf{u}_t \quad \mathbf{u}_{t+1} \quad \dots \quad \mathbf{u}_{t+N-1}], \\ & \mathbf{u}_i \in U_{adm} \quad \forall \quad i = t, t+1, \dots, t+N-1, \\ & \mathbf{x}_i \in X_{adm} \quad \forall \quad i = t, t+1, \dots, t+N. \end{aligned} \tag{4.1}$$

Thus, the objective function of the general MPC-problem consists of a final cost depending on the final state $\mathbf{x}_{t+N}$ and a running cost depending on all states and controls for $i = t, \dots, t+N-1$. The constraints are the system dynamics, the initial condition on the state, $\mathbf{x}_0$ and boundaries for

the control input and states (U_{adm} and X_{adm} are the sets of admissible $\mathbf{u}$ and $\mathbf{x}$). The parameter N is the prediction horizon over which the problem is optimized.

Alternatively, one can let the prediction horizon N tend to infinity and then the objective function of (4.1) is replaced by a convergent sum according to

$$J(\mathbf{x}_t, \mathbf{U}) = \sum_{i=t}^{\infty} \Phi(\mathbf{x}_i, \mathbf{u}_i), \quad (4.2)$$

and thus there is no final cost to consider.

At every sampling time t the following steps are executed:

1. The state $\mathbf{x}_t$ is measured
2. Based on the initial state the objective function is minimized w.r.t $\mathbf{u}_t, \mathbf{u}_{t+1}, \dots, \mathbf{u}_{t+N-1}$.
3. The first control signal $\mathbf{u}_t$ is implemented
4. At the next time instance $t + 1$ the process is repeated from step 1.

4.1 Linear Quadratic MPC

In the case of linear quadratic MPC (LQMPC) the state dynamics are linear and the objective function is quadratic (possibly with linear terms). For a linear time-invariant system the dynamics are given by

$$\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t + \mathbf{B}\mathbf{u}_t, \quad (4.3)$$

and the output

$$\mathbf{y}_t = \mathbf{C}\mathbf{x}_t + \mathbf{D}\mathbf{u}_t. \quad (4.4)$$

Infinite Horizon LQMPC (LQR)

If the prediction horizon tends to infinity and in the absence of linear terms, the objective function becomes

$$J(\mathbf{x}_t, \mathbf{U}) = \sum_{i=t}^{\infty} \mathbf{x}_i^T \mathbf{Q} \mathbf{x}_i + \mathbf{u}_i^T \mathbf{R} \mathbf{u}_i. \quad (4.5)$$

Unfortunately, for the infinite horizon LQMPC problem there only exist a closed form solution in the special case when there are no constraints. Then the optimal control is a linear state feedback

$$u_i = -Kx, \quad (4.6)$$

$$K = (R + B^T P B)^{-1} B^T P A, \quad (4.7)$$

where P is the unique positive definite solution to the discrete algebraic Riccati equation

$$P = A^T P A - A^T P B (R + B^T P B)^{-1} B^T P A + Q. \quad (4.8)$$

This control strategy is commonly known as the linear quadratic regulator (LQR), and is found in [10].

Finite Horizon Constrained LQMP

With a finite horizon and in the absence of linear terms, the objective function becomes

$$J(\mathbf{x}_t, \mathbf{U}) = \mathbf{x}_{t+N}^T \mathbf{P} \mathbf{x}_{t+N} + \sum_{i=t}^{t+N-1} \mathbf{x}_i^T \mathbf{Q} \mathbf{x}_i + \mathbf{u}_i^T \mathbf{R} \mathbf{u}_i, \quad (4.9)$$

with constraints on both control signals and states according to

$$\begin{aligned} \bar{\epsilon}_u \mathbf{u}_i &\leq \bar{\mathcal{F}}_u, \\ -\underline{\epsilon}_u \mathbf{u}_i &\leq -\underline{\mathcal{F}}_u, \\ \bar{\epsilon}_x \mathbf{x}_i &\leq \bar{\mathcal{F}}_x, \\ -\underline{\epsilon}_x \mathbf{x}_i &\leq -\underline{\mathcal{F}}_x, \end{aligned} \quad (4.10)$$

where $\mathbf{Q} = \mathbf{C}^T \mathbf{C}$ and $\mathbf{P}$ and $\mathbf{R}$ are assumed to be positive semi definite diagonal matrices. In this report $\mathbf{P} = \mathbf{Q}$, and then (4.9) can be written on vector form according to

$$J(\mathbf{x}_t, \mathbf{U}) = \mathbf{X}^T \tilde{\mathbf{Q}} \mathbf{X} + \mathbf{U}^T \tilde{\mathbf{R}} \mathbf{U}, \quad (4.11)$$

where

$$\tilde{\mathbf{Q}} = \text{diag}(\mathbf{Q}, \dots, \mathbf{Q}), \quad (4.12)$$

$$\tilde{\mathbf{R}} = \text{diag}(\mathbf{R}, \dots, \mathbf{R}), \quad (4.13)$$

$$\mathbf{X} = (\mathbf{x}_t^T, \mathbf{x}_{t+1}^T, \dots, \mathbf{x}_{t+N-1}^T)^T, \quad (4.14)$$

$$\mathbf{U} = (\mathbf{u}_t^T, \mathbf{u}_{t+1}^T, \dots, \mathbf{u}_{t+N-1}^T)^T. \quad (4.15)$$

The constraints on the control signal and the states are linear

$$\begin{aligned} \epsilon_u \mathbf{U} &\leq \mathcal{F}_u, \\ \epsilon_x \mathbf{X} &\leq \mathcal{F}_x, \end{aligned} \quad (4.16)$$

where

$$\begin{aligned} \epsilon_{\mathbf{u}} &= \begin{bmatrix} \bar{\epsilon}_u & 0 & \dots & 0 \\ 0 & \bar{\epsilon}_u & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \dots & 0 & \bar{\epsilon}_u \\ -\underline{\epsilon}_u & 0 & \dots & 0 \\ 0 & -\underline{\epsilon}_u & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \dots & 0 & -\underline{\epsilon}_u \end{bmatrix}, \quad \epsilon_{\mathbf{x}} = \begin{bmatrix} \bar{\epsilon}_x & 0 & \dots & 0 \\ 0 & \bar{\epsilon}_x & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \dots & 0 & \bar{\epsilon}_x \\ -\underline{\epsilon}_x & 0 & \dots & 0 \\ 0 & -\underline{\epsilon}_x & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \dots & 0 & -\underline{\epsilon}_x \end{bmatrix}, \\ \mathcal{F}_u &= [\bar{\mathcal{F}}_u, \bar{\mathcal{F}}_u, \dots, \bar{\mathcal{F}}_u, -\underline{\mathcal{F}}_u, -\underline{\mathcal{F}}_u, \dots, -\underline{\mathcal{F}}_u]^T, \\ \mathcal{F}_x &= [\bar{\mathcal{F}}_x, \bar{\mathcal{F}}_x, \dots, \bar{\mathcal{F}}_x, -\underline{\mathcal{F}}_x, -\underline{\mathcal{F}}_x, \dots, -\underline{\mathcal{F}}_x]^T. \end{aligned} \quad (4.17)$$

As the dynamics are given by (4.3), all future states $\mathbf{x}_{t+1}, \mathbf{x}_{t+2}, \dots, \mathbf{x}_{t+N-1}$ can be expressed in terms of the current state $\mathbf{x}_t$ and the current and $N-1$ future controls $\mathbf{u}_t, \mathbf{u}_{t+1}, \dots, \mathbf{u}_{t+N-1}$. The objective function (4.11) becomes

$$J(\mathbf{x}_t, \mathbf{U}) = \mathbf{U}^T (\mathcal{B}^T \mathcal{Q} \mathcal{B} + \mathcal{R}) \mathbf{U} + 2\mathbf{U}^T \mathcal{B}^T \mathcal{Q} \mathcal{A} \mathbf{x}_t, \quad (4.18)$$

where

$$\mathcal{B} = \begin{bmatrix} 0 & 0 & \dots & \dots & 0 \\ B & 0 & & & \vdots \\ AB & B & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ A^{N-2}B & \dots & AB & B & 0 \end{bmatrix}, \quad (4.19)$$

$$\mathcal{A} = \begin{bmatrix} I \\ A \\ A^2 \\ \vdots \\ A^{N-1} \end{bmatrix}. \quad (4.20)$$

Minimizing (4.18) subject to (4.16) is a convex optimization problem. In this report the built in Matlab-function quadprog with an interior point method is used, and the method of solving a constrained quadratic programming problem is omitted. This control strategy is often referred to as the standard LQMPC, and is found in [10].

Setpoint Tracking

If the origin is an equilibrium point, minimizing (4.9) is to drive the output to zero. This is not desirable as the objective of the autopilot is to follow a

reference signal given by the guidance law, which almost always is not equal to zero.

An output setpoint $\mathbf{y}_{ref}$ implies that, for a minimal realization of a discrete linear system, the corresponding control signal $\mathbf{u}_{ref}$ and state vector $\mathbf{x}_{ref}$ is uniquely given by

$$\mathbf{u}_{ref} = (\mathbf{C}(\mathbf{I} - \mathbf{A})^{-1}\mathbf{B})^{-1} \mathbf{y}_{ref}, \quad (4.21)$$

$$\mathbf{x}_{ref} = (\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{u}_{ref}, \quad (4.22)$$

in steady state.

As the system is linear one can replace $\mathbf{x}$ and $\mathbf{u}$ in (4.9) by $\mathbf{x}^\delta$ and $\mathbf{u}^\delta$, where

$$\mathbf{x}^\delta = \mathbf{x} - \mathbf{x}_{ref}, \quad (4.23)$$

$$\mathbf{u}^\delta = \mathbf{u} - \mathbf{u}_{ref}. \quad (4.24)$$

Also, the constraints in (4.10) have to be modified such that

$$\bar{\epsilon}_u \mathbf{u}_i^\delta \leq \bar{\mathcal{F}}_u - \bar{\epsilon}_u \mathbf{u}_{ref}, \quad (4.25)$$

etc.

Then, instead of applying the solution from the LQMPC one uses

$$\mathbf{u} = \mathbf{u}^\delta + \mathbf{u}_{ref}, \quad (4.26)$$

and the output $\mathbf{y}$ is thus driven to the reference output $\mathbf{y}_{ref}$.

Integral Action

Linearized models are designed to work in the vicinity of an operating point. If, for example a linearization around the origin is used, and the input signal is a step, there are no guarantees that the steady state error $\rightarrow 0$ as $t \rightarrow \infty$. In fact, for most applications the steady state error is not equal to zero.

In order for the output to converge to the reference signal, integral action has to be introduced. There are several ways to achieve this. With MPC it is particularly easy. If the output of the actual process can be measured, as in the case of the missile, the difference between the output of the model $\mathbf{y}_m$ and the process $\mathbf{y}_p$ gives the offset as

$$\mathbf{D}_{offset} = \mathbf{y}_p - \mathbf{y}_m. \quad (4.27)$$

By inserting $\mathbf{y}_{ref} + \mathbf{D}_{offset}$ instead of the reference output $\mathbf{y}_{ref}$ in (4.21) the steady state error is eliminated.

4.2 Robust MPC

The theory of MPC lacks rigorous stability boundaries. Different combinations of prediction horizons N and weighting matrices $\mathbf{Q}$ and $\mathbf{R}$ may yield large differences in terms of stability, and there is no analytical way to ensure stability without simulating the system with the controller [11]. To address this problem algorithms that are more robust to process/measurement noise and model uncertainties have been suggested. In this report an MPC-algorithm that is robust to modelling errors [12] is implemented.

4.2.1 Model Uncertainties

Consider the linear model in (4.3). If the matrices $\mathbf{A}$ and $\mathbf{B}$ are not known exactly but known to lie within some convex set completely defined by its vertices, $(\mathbf{A} \ \mathbf{B}) \in (\mathbb{A} \ \mathbb{B})$, then the worst case output (worst case w.r.t. the constraints) must correspond to some combination of the vertices [12]. The opposite is not true however, and thus to determine the worst possible outcome all realizations of all possible combinations of $\mathbf{A}$ and $\mathbf{B}$ corresponding to the vertices have to be computed.

For example, if the prediction horizon $N = 10$ and $\mathbf{A}$ and $\mathbf{B}$ belong to the compact set defined by a convex hull with 4 vertices

$$(\mathbb{A} \ \mathbb{B}) = co \{ \{ \mathbf{A}^{(1)} \ \mathbf{B}^{(1)} \}, \dots, \{ \mathbf{A}^{(4)} \ \mathbf{B}^{(4)} \} \}, \quad (4.28)$$

then the number of possible realizations are 4^{10} , which is not feasible to compute at every sampling instance.

A more viable approach is to consign all uncertainties to third term in the dynamics according to

$$\mathbf{x}_{i+1} = \mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{u}_i + \mathbf{D}\mathbf{w}_i. \quad (4.29)$$

(4.29) is a promising approach as the linearized model is a good approximation, but if $\mathbf{w}_i$ is the worst case modelling error one can at least theoretically ensure that the problem does not become infeasible.

Now, the matrices $\mathbf{A}$ and $\mathbf{B}$ in (4.29) are used as if they were true. The state can be decomposed into two parts, one certain ($\mathbf{s}_i$) and one uncertain ($\mathbf{e}_i$),

$$\mathbf{x}_i = \mathbf{s}_i + \mathbf{e}_i. \quad (4.30)$$

If the control signal can be decomposed into

$$\mathbf{u}_i = \mathbf{K}\mathbf{x}_i + \mathbf{c}_i \quad (4.31)$$

the dynamics become

$$\begin{aligned} \mathbf{s}_{i+1} &= (\mathbf{A} + \mathbf{BK})\mathbf{s}_i + \mathbf{B}\mathbf{c}_i, & \mathbf{s}_0 &= \mathbf{x}_0, \\ \mathbf{e}_{i+1} &= (\mathbf{A} + \mathbf{BK})\mathbf{e}_i + \mathbf{D}\mathbf{w}_i, & \mathbf{e}_0 &= 0. \end{aligned} \quad (4.32)$$

4.2.2 Dual Mode MPC

Consider the input decomposed according to (4.31) and the objective function (4.9). The robust LQMPC approach treated in this report is

1. Choose $\mathbf{K}$ to be LQR-optimal according to (4.6)-(4.8).
2. Solve the constrained LQMPC-problem for $\mathbf{c}_i$ for $i = 0, 1, \dots, N - 1$.

If the constraints are inactive, then the quadratic programming yields $\mathbf{c}_i = 0 \forall i = t, t + 1, \dots, t + N - 1$.

This principle is called *Dual mode MPC* as two separate control laws are designed, one for the infinite horizon not taking constraints into account and one for the finite horizon ensuring constraints are satisfied. The concept is illustrated in a block diagram in Figure 4.1.

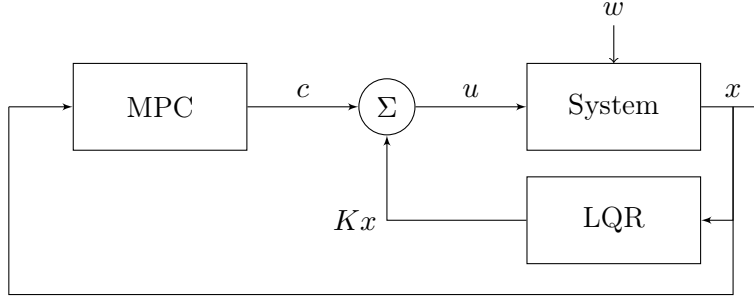


Figure 4.1: Schematic illustration of the dual mode MPC-concept.

Tightened Constraints and Pre-stabilized Predictions

With the state and input constraints given by (4.10) and with the decomposition of states into $\mathbf{s}$ and $\mathbf{e}$ according to (4.32), the constraints for the dual mode MPC-problem can be expressed as

$$\begin{aligned} \mathbf{F}(\mathbf{s}_i + \mathbf{e}_i) &\leq \mathbf{1}, \\ \mathbf{G}(\mathbf{K}(\mathbf{s}_i + \mathbf{e}_i) + \mathbf{c}_i) &\leq \mathbf{1}. \end{aligned} \quad (4.33)$$

As $\mathbf{e}$ evolves according to (4.32) ($\mathbf{e}_0 = 0$), taking the worst case $\mathbf{w}_i$ for each $i = 0, 1, \dots, t + N - 1$ gives tightened constraints. Now, the role of the LQ-regulator of the dual mode becomes apparent, as the feasibility of the problem is ensured through the closed loop predictions.

Maximizing the error, thus taking the worst case scenario, of (4.33) gives

$$\begin{aligned} \mathbf{F}\mathbf{s}_i &\leq \mathbf{1} - \mathbf{F}\mathbf{h}_i, \\ \mathbf{G}(\mathbf{K}\mathbf{s}_i + \mathbf{c}_i) &\leq \mathbf{1} - \mathbf{G}\mathbf{K}\mathbf{h}_i, \end{aligned} \quad (4.34)$$

where

$$\begin{aligned} \mathbf{h}_0 &= 0, \\ \mathbf{h}_i &= \max_{\mathbf{w}_0, \dots, \mathbf{w}_{i-1}} (\mathbf{F} + \mathbf{G}\mathbf{K})\mathbf{e}_i, \quad i = 1, 2, \dots \end{aligned} \tag{4.35}$$

The cost function is the same as the one used to determine the LQR-mode, but now the minimizing argument is $\mathbf{c}_i$. The problem can be expressed on vector form in the same fashion as conventional constrained LQMPC, which then yields a quadratic programming problem on standard form.

Chapter 5

Simulations

In all simulations the International Standard Atmosphere (ISO 2533:1975) is used. The missile has an initial speed of Ma 0.8, which corresponds to the missile being fired from an aircraft. The target is modelled as a point mass with ideal dynamics, meaning that it can change the acceleration instantaneously. The target travels with constant speed.

Two targets are used in the simulation scenarios:

1. A manned enemy fighter aircraft with a maximum admissible acceleration of 9G and a speed of Ma 0.8.
2. A cruise missile with a maximum admissible acceleration of 20G and a speed of Ma 2.

5.1 Simulation Scenarios

4 scenarios are used in the simulations. They are

1. Pursuit with maximum accelerating maneuver.
2. Head on with maximum accelerating maneuver.
3. Pursuit with maximum accelerating weaving maneuver switching every 0.8 s (target acceleration is bang-bang).
4. Head on with maximum accelerating weaving maneuver switching every 0.8 s (target acceleration is bang-bang).

The missiles initial position is $(0, 0, -1000)$ in the NED -frame for all scenarios. For scenario 1 and 3 the enemy aircraft starts in $(2000, 0, -1000)$ and the cruise missile in $(300, 0, -1000)$. For scenario 2 and 4 the aircraft starts in $(5000, 0, -1000)$ and the cruise missile in $(8000, 0, -1000)$.

The simulations are performed with the target maneuver initiated at t_{go} ranging from 0.5 s to 3 s with increments of 0.1 s. Two parameters are of

interest when evaluating the missile performance, namely total flight time and miss distance.

Chapter 6

Results

An example of the missile trajectory is shown in Figure 6.1.

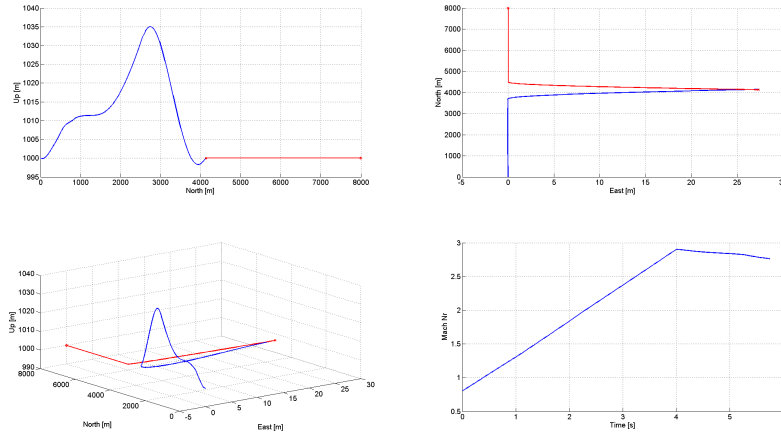


Figure 6.1: The qualitative behaviour of the missile trajectory for scenario 2 with the cruise missile and the target maneuver initiated at $t_{go} = 3$ s. In the bottom right subplot the Mach-number is plotted with time on the x-axis. Note that the scale of the axes or not equal.

6.1 Autopilot

When using the same guidance laws for the different autopilots (conventional LQMPC and robust LQMPC) and with perfect gain scheduling on Ma and altitude (perfect in the sense that the state space model used in the autopilot is updated every sampling time) the two approaches perform equally well. This is because while the 4 scenarios are realistic, the angle of attack/sideslip angle constraints are never active.

The only scenarios when the angle of attack/sideslip angle are active is when the missile makes a large maneuver early in the flight (not treated in this report). The scenarios when the angle of attack/sideslip angle constraints are active always yield large miss distances (> 20 m) for any guidance law, and while the robust LQMPC-algorithm ensures feasibility even when no gain scheduling is used, the same can be achieved with conventional LQMPC by relaxing the constraints. i.e. one can set $|\alpha|, |\beta| \leq 19^\circ$ instead of 20° , and use slack variables with an associated cost a couple of orders of magnitude larger than for the states to ensure the constraints are met [13].

6.2 Guidance Laws

When comparing the guidance laws the conventional constrained LQMPC-law with setpoint tracking and integral action is used.

6.2.1 Aircraft Target

The results for the aircraft target model exhibit the same comparative behavior of the guidance laws as the results obtained for the cruise missile, but with miss distances ≤ 2 m for all the modern guidance laws and scenarios. These results are therefore omitted, as the differences in miss distances are too small for any significant comparison in performance.

6.2.2 Cruise Missile Target

For the cruise missile the differences in miss distance are much larger than for the aircraft, and are seen in Figure 6.2, 6.3, 6.4 and 6.5. The difference in flight times for scenario 2 and 4 were ≤ 0.01 s independent of t_{go} when the target maneuver is initiated. For scenario 1 the flight times vary as seen in Figure 6.6 and for scenario 3 as seen in Figure 6.7. It turns out that for scenario 3 the difference in flight time between PN/DGG and APN/OGL is > 1 s. This is because the target weaving induces a weaving behavior in the missile for APN/OGL, thus causing more induced drag and a longer flight path than for PN/DGG.

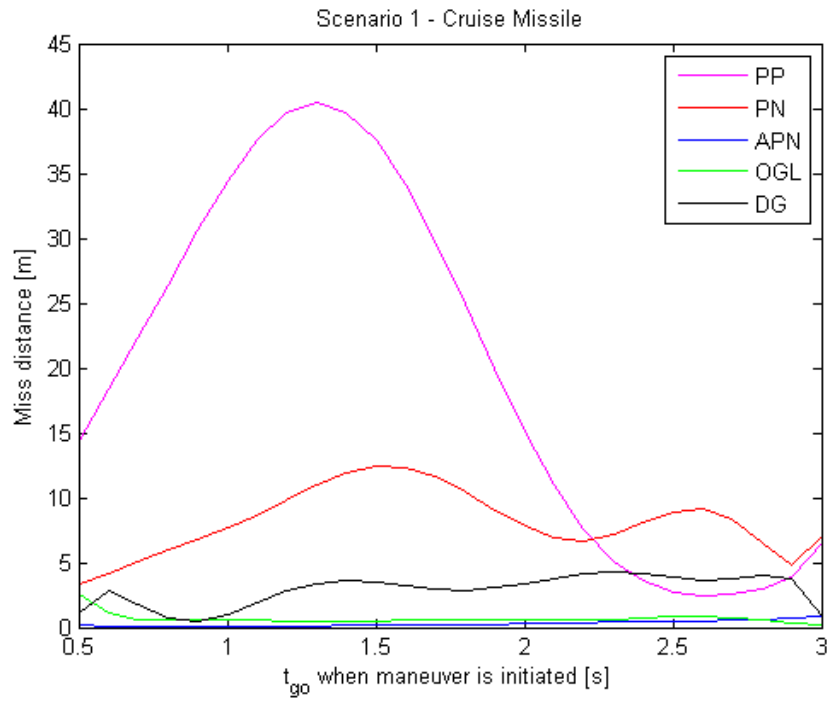


Figure 6.2: Miss distance as a function of the time-to-go when target initiates maneuver in scenario 1 with the target being a cruise missile. (PP stands for "Pure Pursuit", PN for "Proportional Navigation", APN for "Augmented PN", "OGL" for optimal guidance law and "DG" for differential game guidance).

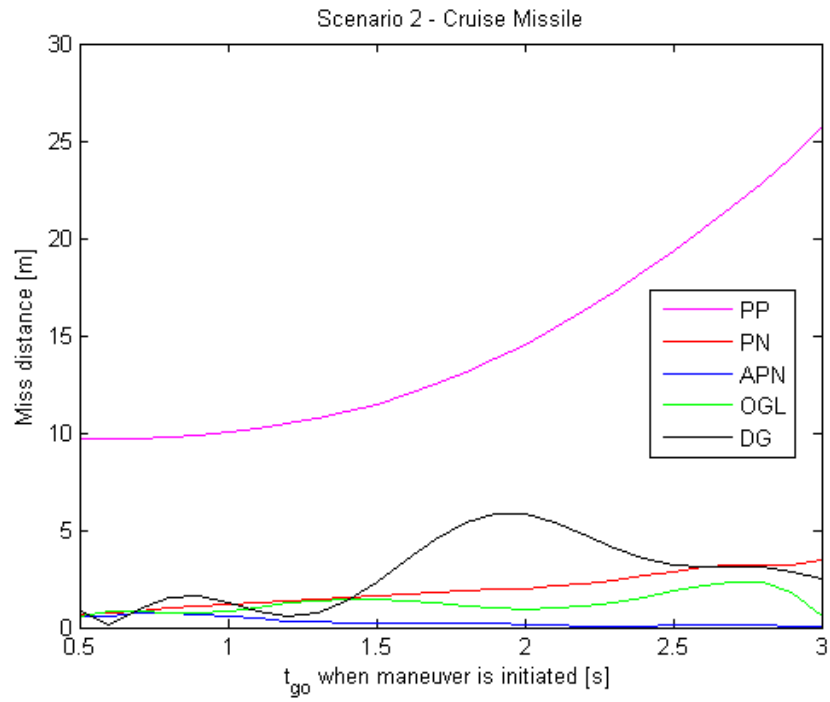


Figure 6.3: Miss distance as a function of the time-to-go when target initiates maneuver in scenario 2 with the target being a cruise missile. (PP stands for "Pure Pursuit", PN for "Proportional Navigation", APN for "Augmented PN", "OGL" for optimal guidance law and "DG" for differential game guidance).

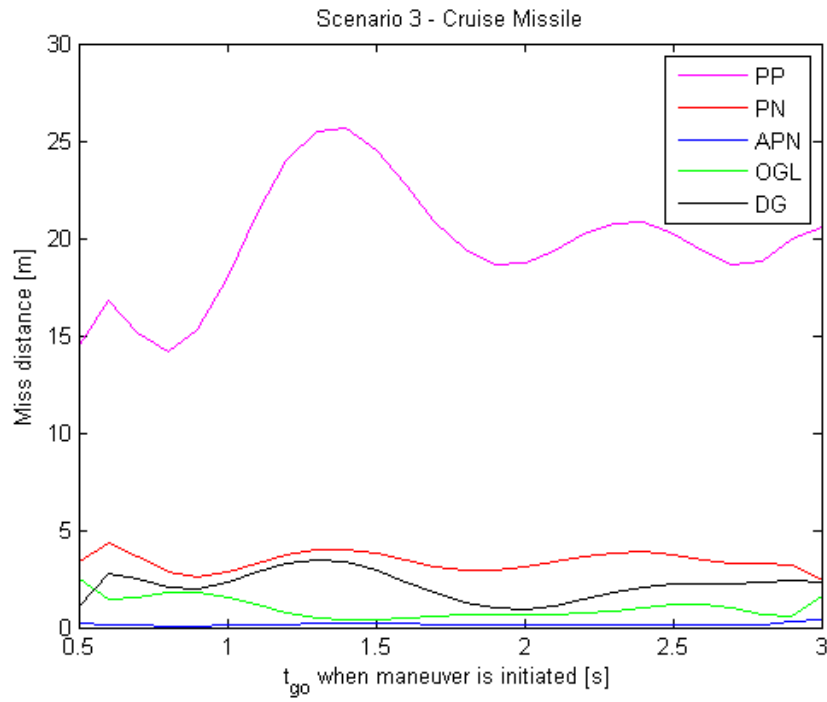


Figure 6.4: Miss distance as a function of the time-to-go when target initiates maneuver in scenario 3 with the target being a cruise missile. (PP stands for "Pure Pursuit", PN for "Proportional Navigation", APN for "Augmented PN", "OGL" for optimal guidance law and "DG" for differential game guidance).

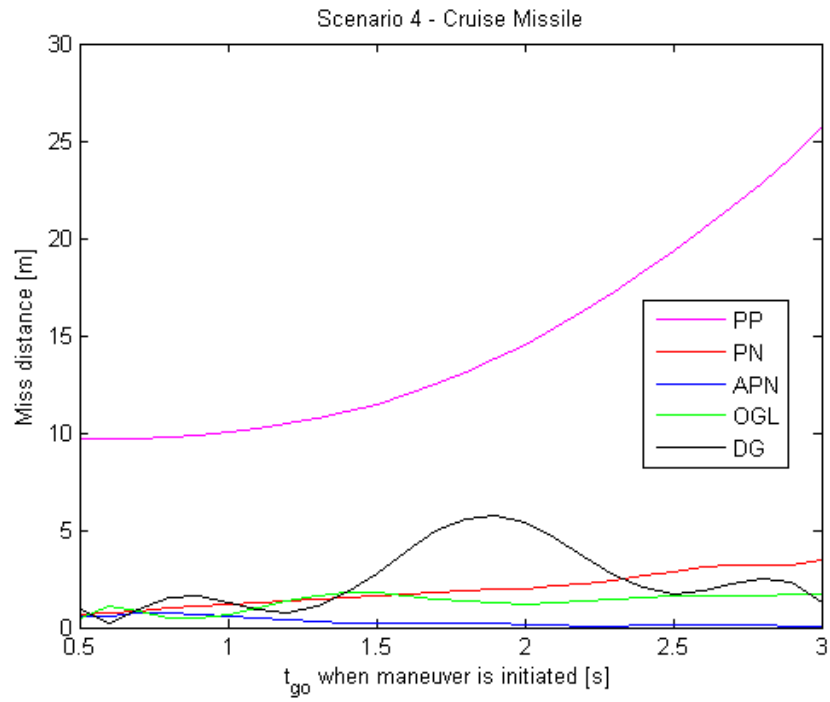


Figure 6.5: Miss distance as a function of the time-to-go when target initiates maneuver in scenario 4 with the target being a cruise missile. (PP stands for "Pure Pursuit", PN for "Proportional Navigation", APN for "Augmented PN", "OGL" for optimal guidance law and "DG" for differential game guidance).

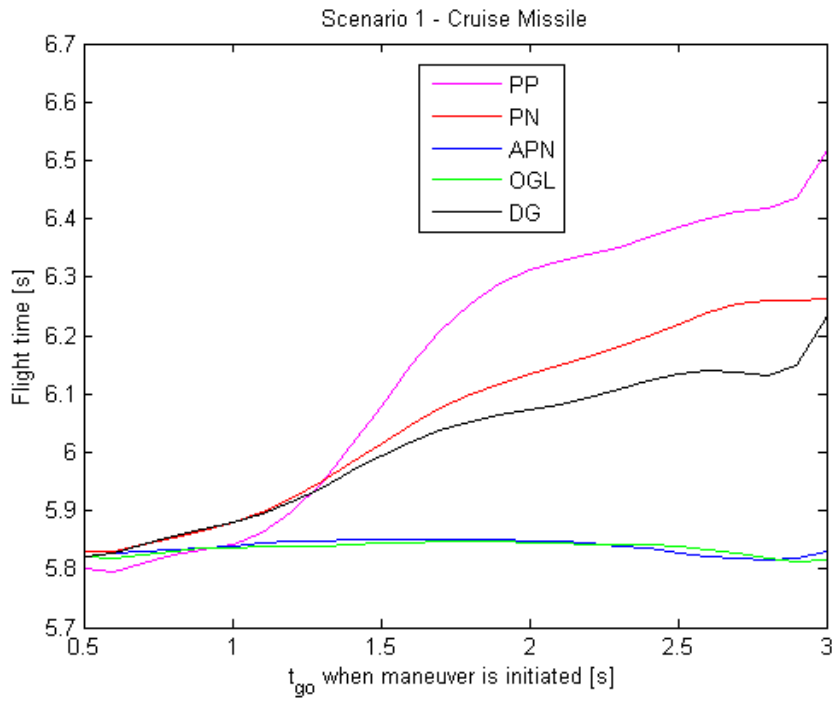


Figure 6.6: Total flight time as a function of the time-to-go when target initiates maneuver in scenario 1 with the target being a cruise missile. (PP stands for "Pure Pursuit", PN for "Proportional Navigation", APN for "Augmented PN", "OGL" for optimal guidance law and "DG" for differential game guidance).

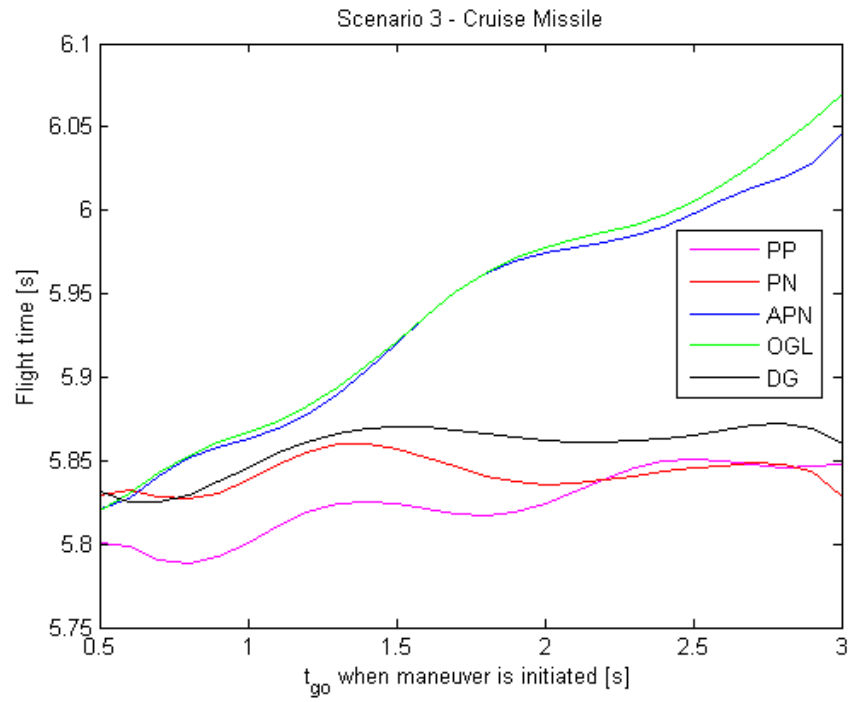


Figure 6.7: Total flight time as a function of the time-to-go when target initiates maneuver in scenario 3 with the target being a cruise missile. (PP stands for "Pure Pursuit", PN for "Proportional Navigation", APN for "Augmented PN", "OGL" for optimal guidance law and "DG" for differential game guidance).

Chapter 7

Discussion, Conclusions and Future Work

7.1 Discussion

The results indicate that for a missile with the purpose to intercept a moving target a maneuverability ratio ($a_{M,max}/a_{T,max}$) advantage of ≥ 2.5 is sufficient for generating miss distances of < 5 m (assuming no process or measurement noise). It is not obvious what is sufficiently small in terms of miss distance to ensure incapacitation of a target, but for the comparison of different guidance laws, the results suffice.

All measurements are assumed to be 100 % accurate, which is never the case in reality. The LOS-angle rate of change is probably not so hard to measure accurately, but the target acceleration perpendicular to the LOS probably is. It would be interesting to see how accurate the data needs to be for APN to be preferred over PN, etc. It is not unlikely to expect a high level of noise/inaccuracy on the measurements of the target acceleration to have an adverse effect on the miss distance performance.

7.2 Conclusions

The missile is able to generate miss distances of < 15 m for all the modern guidance laws even in the case of a cruise missile, therefore it can be concluded that the assumptions/simplifications, from the linearization of the laws of motion to the small angle assumptions in the interception plane are valid.

When intercepting an enemy aircraft (Ma 0.8, 9G) the miss distances are < 2 m for all scenarios, even for PN and DGG. This leads to the conclusion that the limitations in missile control does not lie in the guidance laws or autopilot. This proposition is supported by the fact that in reality, an aircraft trying to evade a missile does not only perform a maneuver, but it does so in

combination with some countermeasure (e.g. flares) to mislead the tracking system of the missile.

Approximating the uncertainty in the model by an additive disturbance turned out to be quite difficult, especially since the aerodynamic coefficients lack analytical expressions. In this case, the feasibility for the cases when the angle of attack/sideslip angle constraints are active can be assured simply by relaxing the constraints, which does not require a very profound understanding of the modelling uncertainties. The conclusion is thus, that the robust MPC used in this report is better used for simpler models, alternatively not at all to handle model uncertainty but rather for process process disturbances.

7.3 Future Work

Based on the discussion in 7.1 it is suggested that the following aspects are considered in future work:

1. The inclusion of measurement and process noise in the model. Here the robust MPC-algorithm presented in 4 could probably be used with greater success, as the effects of process disturbances are easier to capture by a third term in the state space model.
2. Explore the possibility of a hybrid guidance law, utilizing the advantages of the different guidance laws. For example, DGG outperforms PN in the scenarios where the missile and target are initially in pure chase, but for head on collision that is not the case.

This can be incorporated with a target tracking algorithm, based on artificial intelligence, classifying different target maneuvers.

3. Compare different autopilots with respect to computational effort, and then evaluate the trade off between increased computational complexity at the cost of decreased sampling rate from a performance point of view.

Bibliography

- [1] M. Ekström. *Slutfasstyrning av missil med explicit prediktionsreglering*. Master thesis LITH-ISY-EX-3619-2005, Department of Electrical Engineering, Linköping University, Linköping, Sweden, 2005.
- [2] D. Rosdal. *Missilstyrning med Model Predictive Control*. Master thesis LITH-ISY-EX-3608-2005, Department of Electrical Engineering, Linköping University, Linköping, Sweden, 2005.
- [3] H. Johansson. *Gain scheduled missile control using robust loop shaping*. Master thesis LITH-ISY-EX-3291-2003, Department of Electrical Engineering, Linköping University, Linköping, Sweden, 2003.
- [4] P. Zipfel. *Modeling and simulation of aerospace vehicle dynamics*. AIAA education series, American Institute of Aeronautics and Astronautics, 2007.
- [5] B. Wittenmark, K.J. Åström, K.-E. Årzén. *Computer control: An overview*. International Federation of Automatic Control (IFAC), Professional Briefs, 2002.
- [6] P. Zarchan. *Tactical and Strategic Missile Guidance*. American Institute of Aeronautics and Astronautics, 2007.
- [7] L. Råde, B. Westergren. *Mathematics Handbook for Science and Engineering*. Studentlitteratur, 1988.
- [8] R. Isaacs. *Differential games : a mathematical theory with applications to warfare and pursuit, control and optimization*. John Wiley and Sons, Inc., 1965.
- [9] S.N. Balakrishnan, A. Tsourdos, B.A. White. *Advances in Missile Guidance, Control and Estimation*. CRC Press, 2012.
- [10] T. Glad, L. Ljung. *Reglerteori*. Studentlitteratur, 1997.
- [11] J. Löfberg. *Minmax approaches to robust model predictive control*. Ph.D. thesis ISBN 91-7373-622-8, Department of Electrical Engineering, Linköping University, Linköping, Sweden, 2003.

- [12] B Kouvaritakis, M. Cannon. *Model Predictive Control - Classical Robust and Stochastic*. Springer, 2015.
- [13] P.O.M. Scokaert, J.B. Rawlings. *Feasibility Issues in Linear Model Predictive Control*. AIChE Journal, 1999.

Appendices

Appendix A

Plots of Coefficients

In this chapter the plots illustrating the relationship between Ma , α , β , δ_a , δ_e , δ_r and the coefficients CA , CY , CN , Cm , Cn , Cl are presented.

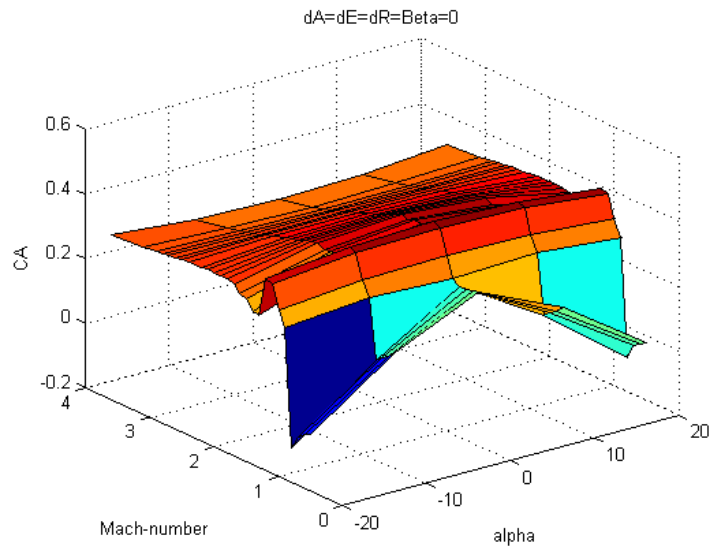


Figure A.1: CA as a function of Mach-number and angle of attack α ($\beta = \delta_a = \delta_e = \delta_r = 0$). The corresponding plot for β is identical.

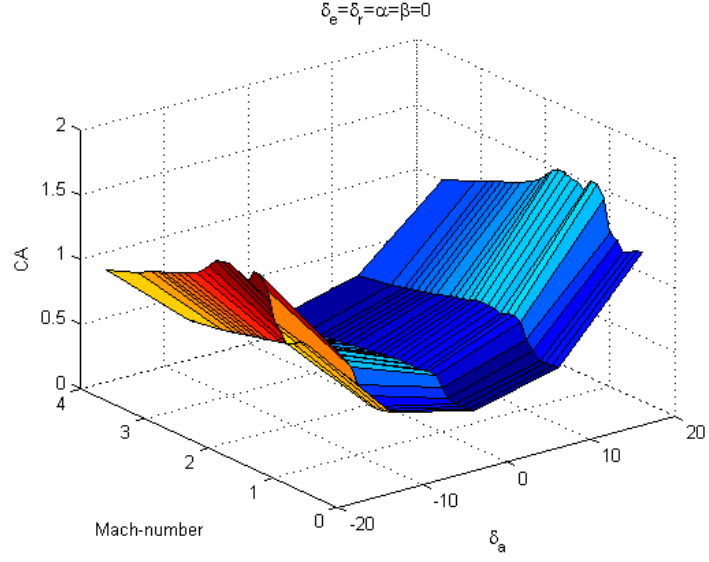


Figure A.2: CA as a function of Mach-number and aileron deflection angle δ_a ($\alpha = \beta = \delta_e = \delta_r = 0$). The corresponding plots for δ_r and δ_e are identical.

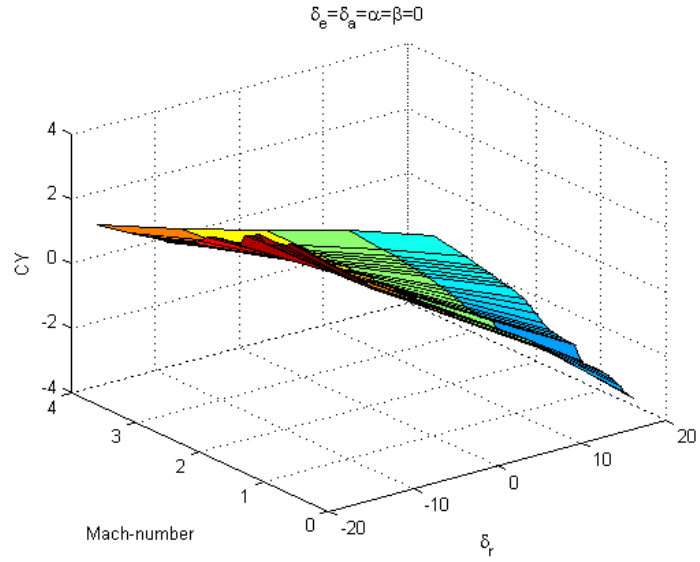


Figure A.3: CY as a function of Mach-number and rudder deflection angle δ_r ($\alpha = \beta = \delta_e = \delta_a = 0$). The corresponding plots for δ_a and δ_e are zero. The same graph is obtained from plotting CN with respect to δ_e instead.

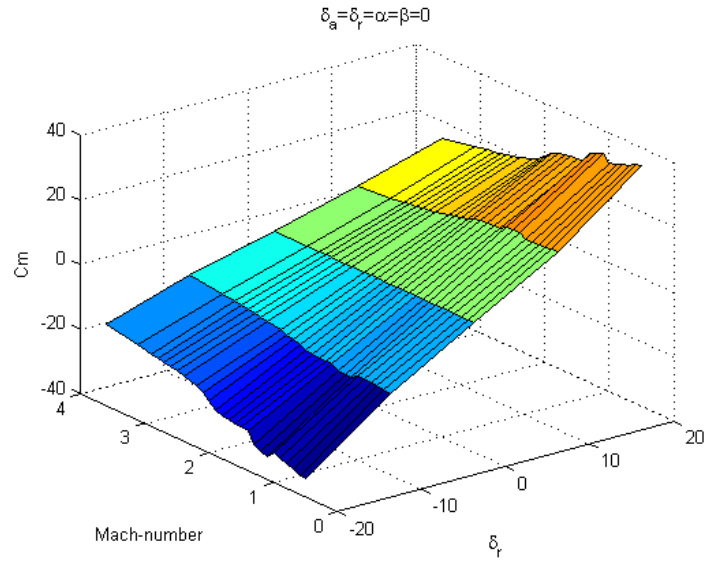


Figure A.4: C_m as a function of Mach-number and elevator deflection angle δ_e ($\alpha = \beta = \delta_r = \delta_a = 0$). The same graph is obtained from plotting C_n with respect to δ_r instead.

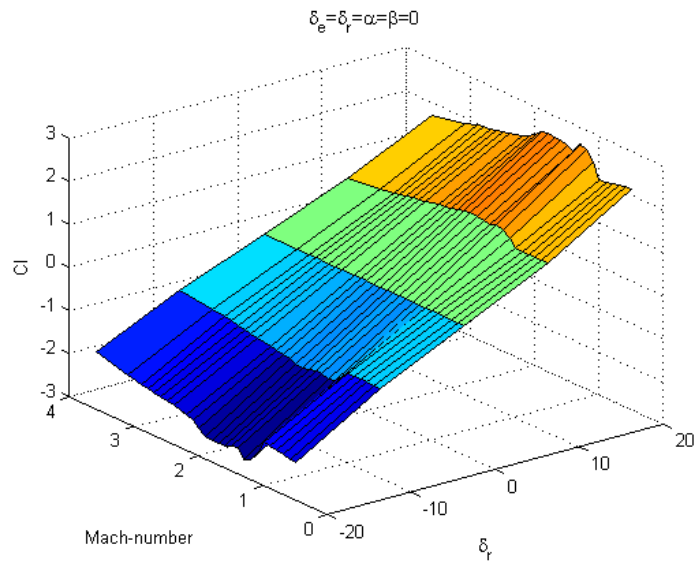


Figure A.5: Cl as a function of Mach-number and aileron deflection angle δ_a ($\alpha = \beta = \delta_r = \delta_e = 0$).

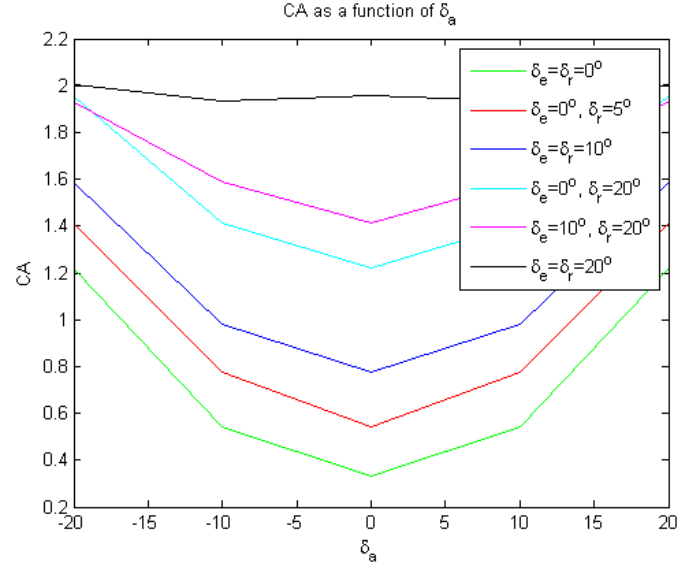


Figure A.6: CA as a function of aileron deflection angle δ_a ($\alpha = \beta = 0$ and various combinations of δ_e and δ_r). The corresponding plots for δ_r and δ_e are identical.

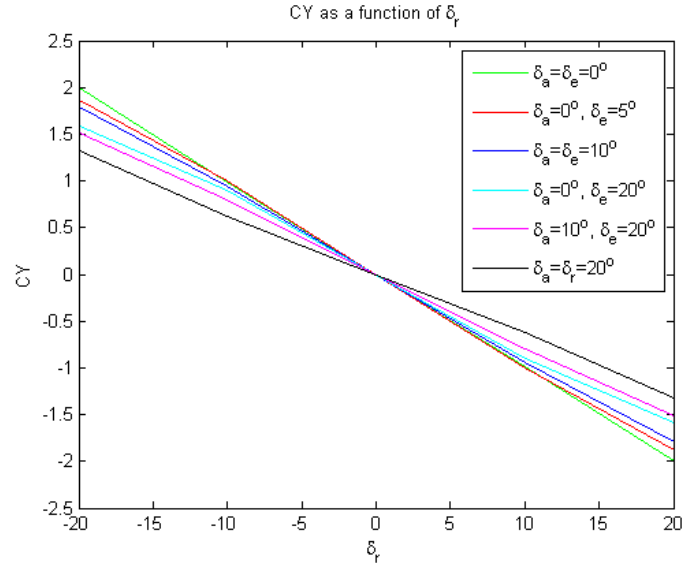


Figure A.7: CY as a function of rudder deflection angle δ_r ($\alpha = \beta = 0$ and various combinations of δ_a and δ_e).

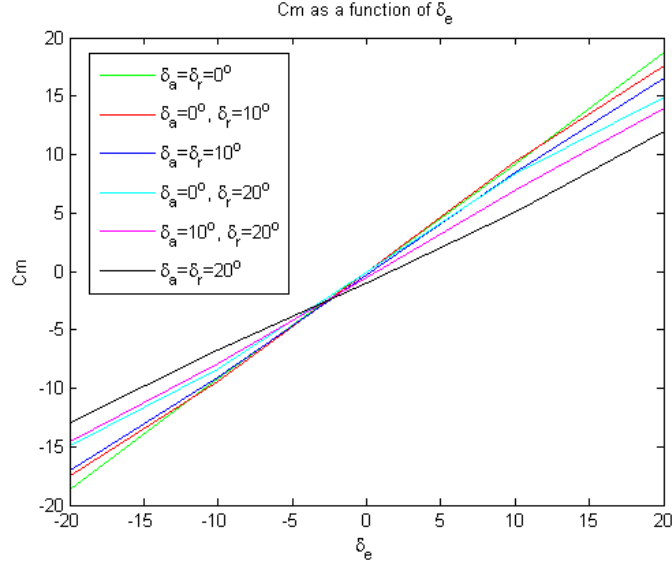


Figure A.8: C_m as a function of elevator deflection angle δ_e ($\alpha = \beta = 0$ and various combinations of δ_a and δ_r).

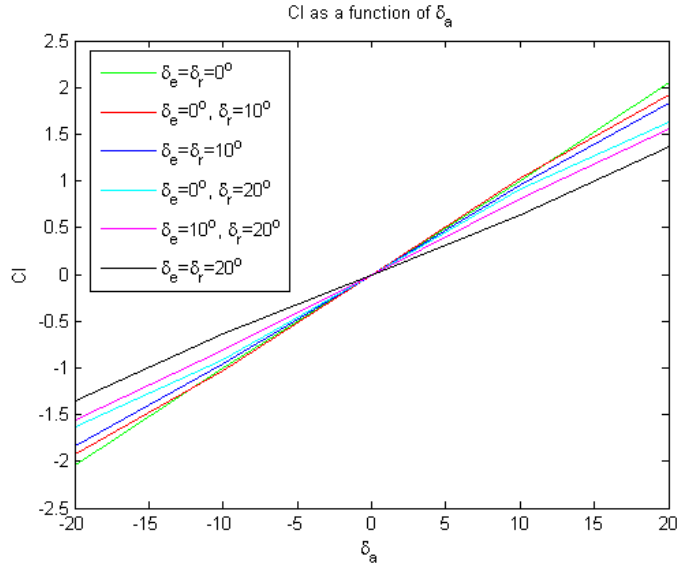


Figure A.9: C_l as a function of elevator deflection angle δ_a ($\alpha = \beta = 0$ and various combinations of δ_e and δ_r).

TRITA -MAT-E 2016:38
ISRN -KTH/MAT/E--16/38--SE